17 Eastern English Channel sole

sol.27.7d – *Solea solea* in Division 7.d

This section of the report provides a comprehensive description of the methods and data used for the 2025 assessment of sole in Division 27.7.d. Additional background information can be found in the Stock Annex. This year's assessment shows that the recruitment estimates of the years 2023 and 2024 are highly uncertain due to a lack of data that informs the dynamics of the age 1 individuals, as well as missing survey data from the French Young Fish Survey for 2024. These recruitment estimates may be biased with knock-on effects on the short term forecast and advice. However, the uncertainty is accounted for in the forecasted recruitment by weighting the resampled recruitment with the associated standard deviation.

17.1 General

17.1.1 Stock definition

During the WKNSEA 2017 and WKNSEA 2021 benchmark, the available information on stock identity was investigated, including genetic, tagging and otolith information. Sole in the eastern English Channel (7.d) is still considered to be a stock separated from the larger North Sea stock (27.4) to the east and the smaller geographically-separated stock to the west in 27.7.e (western English Channel). Considering the sub-stock structure, three regions with low connectivity were identified within Division 7.d for larvae, juveniles and adults (Archambault *et al.*, 2016; Lecomte *et al.*, 2020; Randon *et al.*, 2018; 2020). More information is provided in the Stock Annex, the report of the benchmark and the associated working document (ICES, 2021).

17.1.2 Ecosystem aspects

A general description of the available information on ecological aspects can be found in the Stock Annex.

17.1.3 Fisheries

A general description of the fishery is presented in the Stock Annex.

17.1.4 Management regulations

Management of sole in Division 27.7.d is by TAC and technical measures.

The Minimum Conservation Reference Size for sole is 24 cm (EU legislation). Sole in the eastern English Channel is fully under the landing obligation since 2018 (partially since 2016) (EU, 2018/2034). There are two exemptions in place which allow for discarding of undersized sole in Division 7.d:

1) a survival exemption for small coastal otter trawlers (<10 m and <221 kW) fishing less than 90 minutes in areas with a depth less than 30 m (outside nursery areas) and with cod-end mesh size of 80–99 mm.

2) a *de minimis* exemption for vessels using trammel and gill nets (max. 3% of annual catches) and using TBB gear with a mesh size of 80–119 mm equipped with the Flemish panel (max. 3% of annual catches).

A historical overview of the TAC for sole 7.d since 2014 is presented in the table below (short version) and in Table 17.1.

Table 17.1. Historical overview of the TACs for sole in Division 27.7.d (2014–2025); Note: TAC represents catch from 2016 onwards (landing obligation)

Year	2014	2015	2016	2017	2018	2019
TAC	4838	3483	3258	2724	3405	2515
Year	**2020**	**2021**	**2022**	**2023**	**2024**	**2025**
TAC	2797	3248	2380	1747	1504	1209

Since 2010, full uptake of the sole 27.7.d TAC has not been realized. However, over the last 3 years (2022 to 2024), the initial country-specific TAC share became restrictive for Belgium, and in 2015 this was the case for Belgium and France (see 17.2.1.1 TAC uptake). In 2024, ICES estimated catches were higher than the TAC share. This was mainly driven by Belgium and France. Note that initial quota are compared regardless of quota exchanges among countries.

In response to the drop in spawning stock biomass (SSB) and the poorer recruitment from 2012 onwards, the main countries participating in the fishery implemented additional conservation measures. For Belgian beam trawlers in 27.7.d (and 27.7.fg, 27.7.a), it is mandatory since 1 April 2015 to incorporate a 3 m long section (tunnel) with a 120 mm mesh size before the cod-end (Flemish panel), in order to reduce the catches of small sole (reduction of undersized sole with 40% and marketable sole with 16%). France engaged in 2016 to i) strengthen the protection of the nursery areas, ii) increase the area closed to fishing within the nursery areas, and iii) increase the minimum conservation reference size to 25 cm for French vessels in accordance with EU legislation, where appropriate. From 11 March until 31 December 2017, the minimum conservation reference size for Belgian vessels also increased to 25 cm. This MCRS was used up to 14 February 2025, but was since the 15th of February 2025 changed back to 24 cm. Finally, UK beam trawlers usually fish using mesh sizes greater than statutory in order to avoid discarding and to avoid wasting quota.

17.1.4.1 Additional information provided by the fishing industry

In 2019, the French fishing industry provided input on their perceived status of the stock.

The French gillnet fishers state that they have trouble catching sole in the eastern part of the eastern English Channel. The French otter trawl fishers operating mainly in the south-western part of the eastern English Channel have reported a decline in catches in 2016 and 2017, followed by an increase in catches since 2018 to the ten-year average level.

17.1.5 ICES advice

17.1.5.1 ICES advice for 2024

The ICES advice for 2024 was:

ICES advises that when the MSY approach is applied, catches in 2024 should be no more than 1504 tonnes.

In 2023, the stock status was presented as follows:

	Fishing pressure				Stock size			
		2020	2021	2022		2021	2022	2023
Maximum sustainable yield	F_{MSY}	✗	✓	✗ Above	MSY $B_{trigger}$	✗	✗	✗ Below trigger
Precautionary approach	F_{pa},F_{lim}	✓	✓	✓ Harvested sustainably	B_{pa},B_{lim}	○	✗	○ Increased risk
Management plan	F_{MGT}	✓	✓	✓ Above	B_{MGT}	✗	✗	✗ Increased risk

17.1.5.2 ICES advice for 2025

The ICES advice for 2025 was:

ICES advises that when the MSY approach is applied, catches in 2025 should be no more than 1209 tonnes.

In 2024, the stock status was presented as follows:

	Fishing pressure				Stock size			
		2021	2022	2023		2022	2023	2024
Maximum sustainable yield	F_{MSY}	✓	✗	✓ Below	MSY $B_{trigger}$	✗	✗	✗ Below trigger
Precautionary approach	F_{pa},F_{lim}	✓	✓	✓ Harvested sustainably	B_{pa},B_{lim}	✗	✗	✗ Reduced reproductive capacity
Management plan	F_{MGT}	✓	✗	✓ Below	B_{MGT}	✗	✗	✗ Below

17.2 Data

17.2.1 Catches

17.2.1.1 TAC uptake

Table 17.1 and Figure 17.1 summarise the official sole landings by country for Division 27.7.d. The landings have steadily increased over the 1970s and 1990s, fluctuated around an average of 4838 t in 2000–2014 (range: 3832 t–6247 t), and dropped to 3411 tonnes in 2015 and even further to 2224 tonnes in 2017. In 2018, a small increase up to 2312 tonnes was observed. In the period 2019-2022, the landings stabilized around 1700 tonnes, but decreased further to 1067 tonnes in 2024. Since 2000, the contribution to the landings of the three main countries involved in this fishery has remained stable over time (~30% Belgium, ~15% UK, and ~55% France) (Figure 17.2). However, since 2019, the proportion of landings caught by the Belgian fleet has increased, while landings by the French fleet have decreased. In 2024, the Belgian fleet caught 39%, the French fleet 49% and the UK fleet 13% of sole landings in Division 7.d.

Since 2010, full uptake of the sole 27.7.d TAC has not been realized. However, in 2014, the Belgian initial quota was overshot by 15%. In 2015, Belgium overshot its national, initial quota again by 12% and France faced a 1% overshoot. The total uptake in 2015 was 98% (official landings). Since then, the total uptake has decreased to only 51% in 2021. Nevertheless, the strong reduction of the TAC in 2022, 2023 and resulted in an increase of the quota uptake, with an overshoot (3% in 2022, 18% in 2023 and 5% in 2024) of the initial quota by Belgium. Note that the initial quota is compared with uptake not taking into account quota exchange among countries during the year.

When comparing ICES catch estimates (InterCatch) with the TAC (catch), a total uptake of 70% was realised in 2020, 59% in 2021, 85% in 2022, 87% in 2023 and 105% in 2024 (Figure 17.3). Figure 17.4 presents a historic overview of TAC levels compared to official landings and ICES estimates (both landings and discards).

17.2.1.2 ICES catch estimates (InterCatch)

New ICES estimates were uploaded and processed in InterCatch from 2004 onwards as a result of the WKNSEA 2021 benchmark. The new upload involved a thorough revision of the French and Belgian time-series (more information in the WKNSEA 2021 benchmark report: ICES, 2021). In 2025, UK England updated its data for 2021 to 2023. Changes were minimal: less than 1% change in InterCatch landings and discards in 2021, a change of 11 and 5 tonnes in InterCatch landings, and 17 and 18 tonnes in InterCatch discards, in 2022 and 2023, respectively.

The proportion of landings with associated discards has gradually increased over the years 2004–2012 (Figure 17.5). From 2012 onwards, this increasing trend levelled off, fluctuating between 70 and 80%. A decrease was noted in 2020 to 54%, most likely due to the Covid-19 pandemic. In 2024, the proportion of landings with discards was 73%. The age coverage for landings increased from 2004–2011 and remained stable around 80% (Figure 17.5). The age coverage for 2024 is 86% and shown by country and by fleet in Figure 17.6 and 17.7 respectively. The age coverage for discards fluctuates around 60% over the whole time-series and is at 73% in 2024 (Figure 17.5).

A detailed overview of imported or raised data and sampled or estimated distributions for 2024 is given in Table 17.2.

Discards are included in the assessment from working year 2017 onwards. If discards are unavailable for a particular year-quarter-country-métier combination, they are assumed to be unknown (non-zero) and therefore raised (InterCatch). The weighting factor for raising the discards was '*Landings CATON*' (landings catch).

Discard raising was performed on a **gear level** regardless of season or country. The following groups were distinguished based on gear:

- TBB
- OTB including OTB, OTT, SSC, SDN
- GTR including GTR and GNS

The remaining gears were combined in a REST group (including MIS, FPO, DRB, LHM, LLS).

The GNS/GTR, TBB and OTB/OTT/SSC/SDN contribute respectively 30%, 43% and 21% to the landings of sole in 27.7.d (Table 17.3).

Raising within a gear group was performed when the proportion of landings for which discard weights are available was **equal or larger than 50%** compared to the total landings of that group. For the 2021 and 2022 data, this was the case for the TBB, GTR and OTB gear group. In 2023 and 2024, this threshold was only reached for the OTB and TBB group. The remaining gears were grouped in the REST group, which was raised using all available information except for strata with a discard rate larger than 50%.

To **allocate age** compositions, landings and discards were handled separately; samples from landings were used only for landings and *vice versa*. When age distributions (both landings and discards) had to be borrowed from other strata, allocations were performed on a **gear level**. The same gear groups (TBB, OTB, GTR and REST) as used for discard raising were applied. When the **threshold of 50%** was reached for the proportion of landings or discards covered by age, allocation of age occurred with all available information within that gear group. For the landings data from 2021-2024, this threshold was reached for all gear groups. For the 2021 discards, this was the case for the TBB and GTR group and for the 2022 and 2023 discards, this was only the case for the TBB group. For the 2024 discards, this was the case for the TBB and OTB group. When the threshold was not reached, unsampled data were pooled in the REST group and ages were allocated using all sampled data. The weighting factor was '*Mean Weight weighted by numbers at age*'.

From 2018 onwards, **BMS landings** and **logbook registered discards** were available in InterCatch. However, all were zero up to 2020. In 2020, 247 kg of BMS was reported from the English

GNS_DEF_all Q4 and OTB_DEF_70-99 Q4 strata. In 2021, 205 kg of BMS was reported from the English GNS_DEF_all Q1, OTB_DEF_70-99 Q1 and GTR_DEF_all Q3 strata. In 2022, 2023 and 2024, no BMS landings and logbook registered discards were reported. Logbook registered discards were not considered for the age allocations. Age allocation of BMS landings was done together with discards.

The official catch statistics have reported BMS landings in 2017 (144 kg), 2019 (2.8 kg), 2020 (249 kg), 2021 (415 kg), 2022 (835 kg), 2023 (486 kg) and 2024 (339 kg). No BMS landings were reported in 2018.

17.2.1.3 Reconstruction of discards

Due to the lack of discard information prior to 2004, discards were reconstructed for the period 1982–2003 (ICES, 2021). Similarly, as during the WKNSEA 2017 benchmark, an average discard proportion at age was calculated for the period 2004–2008. This decision was motivated by the fact that discard behaviour at age changed after 2008 and a general increase in discarding was found in the most recent years (Figure 17.8).

17.2.1.4 Discard rate

The discard rate, calculated as the ratio between ICES discard estimates (tonnes) and ICES catch estimates (tonnes), fluctuates between 3 and 10% over the time-series (1982–2018) (Figure 17.9). However, in 2019, a sudden increase in the discard rate occurred to approximately 20%, levelling off to 16% in 2022 and 15% in 2023. However, in 2024, a strong increase in discards (30%) is denoted mainly driven by high discard rates of the French OTB_DEF_70-99_0_0 fleet. For a target species such as sole, this recent rate is very high.

In 2020, as a result of the Covid-19 pandemic, hampered sampling could have been affecting the discard rate (see §17.2.1.2). Usually most of the imported discards originate from France (62% in 2018) followed by Belgium (34% in 2018) and England (4% in 2018). However, in 2020, 66% of the imported discards originated from Belgium and only 34% from France. This was the result of the reduced sampling by France due to the pandemic. Belgium only submits discard data from the TBB group, while France usually provides a mix of GTR, OTB and TBB strata. Following the discard raising procedures (§17.2.1.2), the threshold of raising the OTB and GTR group using only OTB and GTR strata respectively was not met for the 2020 data (in contrast to the three preceding years). Consequently, these strata were raised in a REST group including BEL TBB_DEF_70-99 (DR = 0.22), FRA GTR_DEF_100-119 Q1 (DR = 0.023), Q3 (DR = 0.018), Q4 (DR = 0.010), FRA GTR_DEF_90-99 Q1 (DR = 0.044), FRA OTB_DEF_70-99 Q3 (DR = 0.326) and ENG OTB_DEF_70-99 Q1 (DR = 0). The French OTB_DEF_70-99 Q1 stratum was not included because the discard rate exceeded 50% (DR = 0.514). Consequently, the Belgian discard samples had a higher impact for the 2020 data compared to previous years. In 2020, Belgian discard rates were unusually high (22% versus 12% in 2018 and 14% in 2019). However, the Belgian observer programme was not hampered by the Covid-19 pandemic.

Other explanations could be good recruitment entering the fishable population or decreasing size at age. However, it remains currently unclear why the discard rate increased in 2024.

17.2.1.5 Numbers-at-age

Catch, landings and discards numbers-at-age are shown in Figure 17.10, 17.11, 17.12, 17.13 and Table 17.4.

Catch numbers have decreased over the time-series and lower numbers are caught since 2015 (Figure 17.10). In 2008–2009, a stronger year class entered the stock and was found in the landings from age 2 onwards. The 2018-year class is the first since 2008-2009 that seems large enough to be observed in the landings. However, fewer 2-year olds have been observed in the landings compared to 2008-2009.

Almost half of the 2- and 3-year-old fish were discarded in 2020. In contrast to another period with a strong year class (e.g. 2001-year class), it is apparent that proportionally more 2-year olds end up in the discard fraction (Figure 17.10). In 2024, catches comprised mainly age 2 fish in terms of numbers. The high share of age 2 fish in the catches was also reflected in the discards, 62% of the total discard volume were fish of 2 years old.

Additionally, Figure 17.13 shows a considerable amount of older discards, especially in 2019 and 2020. Considering the larger impact of the Belgian samples in 2020, Belgian age-length keys (modelled) were investigated. Discards up to 29 cm were aged and while in 2019 over 40% were younger than age 6, in 2020 this was less than 5%. This confirms the pattern of decreasing length-at-age also found in the UK BTS survey data (ICES, 2021).

17.2.1.6 Weight-at-age

Weights-at-age for discards and landings are shown in Figure 17.14 and 17.15 respectively and weights-at-age in the catch are given in Figure 17.16 and Table 17.5. Catch weights-at-age (Figure 17.16) have gradually declined over the past 10 years, especially for the older ages (age 6+). Furthermore, although discards of older ages are being caught (up to age 8 and older), their mean weight-at-age varied around 100 grams. This points to a trend of decreasing weight-at-age and thus smaller (§17.2.1.5) and thinner fish.

17.2.2 Stock weight-at-age

Stock weights-at-age were revised during the WKNSEA 2021 benchmark (ICES, 2021). Quarter 1 catch weight-at-age are extracted from InterCatch and used as stock weights (Figure 17.17; Table 17.6).

17.2.3 Maturity and natural mortality

During the WKNSEA 2017 benchmark (ICES, 2017), the knife-edged maturity ogive with full maturation from age 3 onwards was revised. Using data from the French IBTS survey and commercial data from Belgium, France and the UK (15 191 records), a new maturity ogive was constructed (see table below). More information on how this was achieved is provided in the WKNSEA 2017 report and the associated working document (ICES, 2017).

Age	0	1	2	3	4	5	6	7	8	9	10	11(+)
Maturity	0.00	0.00	0.53	0.92	0.96	0.97	1.00	1.00	1.00	1.00	1.00	1.00

Natural mortality is assumed constant over ages and years at 0.1. English and French tagging data were investigated in light of estimating natural mortality (prior to the WKNSEA 2021 benchmark), but two problems were encountered. First, most of the tagging data dated back to before the beginning of the sole 7.d time-series. Second, in the most recent years, there were too little recaptures which inhibited the calculation of a new estimate for natural mortality (Lecomte *et al.*, 2020).

17.2.4 Tuning series

The assessment of sole in the eastern English Channel is tuned with three survey (UK(E&W)-BTS-Q3, UK-YFS and FRA-YFS) and three commercial tuning series (FRA-COTB, UK(E&W)-CBT and BE-CBT).

17.2.4.1 Commercial tuning indices

During the WKNSEA 2021 benchmark, the Belgian commercial beam trawl index and the French commercial otter trawl index were revised (ICES, 2021). The UK commercial beam trawl index was revised during the 2019 inter-benchmark (ICES, 2019). A minor change was made to the Belgian commercial beam trawl series during the inter-benchmark in June 2022 (ICES, 2022). This change included the removal of the rectangle threshold on the data. All three commercial tuning fleets are included in the assessment as fishable biomass indices (aggregated over all age) (Figure 17.18; Tables 17.7-9). The Belgian and French index follow the same trend with the exception of the year 2020 and 2022, where the Belgian index increases or stabilises and the French index decreases. The UK commercial index gives information back up to 1986 and shows an opposite trend around 2005 and in 2008–2014 compared to the Belgian and French index. The opposite trends could be explained by the specific area where the UK beam trawl fleet is fishing (along the southern English coasts). Since 2014, trends are similar up until 2020, where the UK index shows an increase while the other indices point to a decrease. In 2024, all commercial indices show a decrease, especially the French index. Note however that for the French OTB fleet, a very high discard rate was reported (see 17.1.2.4). As the TAC declined considerable over the last years, it should also be investigated how consistent the lpue indices are with respect to eventual changes in targeting behavior due to quota limitations.

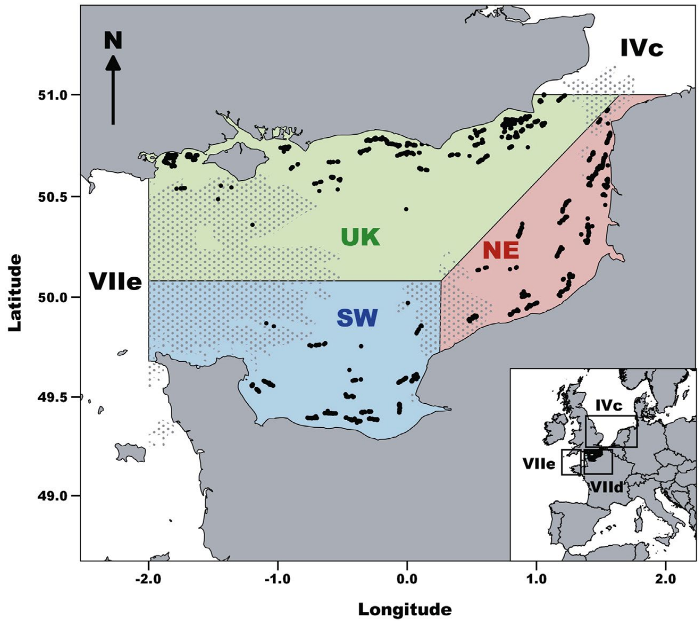
17.2.4.2 Survey tuning indices

The survey tuning fleets are included as age-disaggregated indices. For the UK beam trawl survey (BTS) information from ages 1–6 is available and the UK and French Young fish surveys provide information on age 1 (Figure 17.19; Tables 17.10-12).

During the 2022 IBP, the UK BTS survey was investigated (Table 17.10). It became clear that the survey was able to track the cohorts better in the beginning of the time-series compared to the end of the time-series as shown by Figure 17.20 and 17.21. The internal consistency plots showed that the correlation between ages 1-2, 2-3 and 3-4 changed considerably, while this correlation was more stable for ages 4-5 and 5-6 (Figure 17.21). The age information from ages 1-3 from 2010 onwards was therefore removed. This was most likely related to the changing size at age. However, uncertainty around age readings was also brought up to explain the observed change. This will be investigated further. The UK BTS survey was not conducted in 2022, and therefore, no information was available for 2022.

The French YFS (Table 17.12), which is confined to the Somme estuary (see stock annex), shows a rather constant and low index in the period 2014–2018 (Figure 17.19). In 2019 and 2020, an increase in recruitment is measured. The values for 2021 and 2022 are again back to levels prior to 2019, although the 2023 estimates show a small increase. For 2024, no data point was available. The UK YFS (Table 17.11) stopped in 2006 and was situated along the southern English coast.

During the WKNSEA 2021 benchmark, evidence for the presence of three subpopulations was investigated (Lecomte *et al.*, 2020; Randon *et al.*, 2018, 2020; indicated on map below). The UK BTS data was further analysed and higher abundances of sole were found in the south-western population (Seine Bay), followed by the northern UK subpopulation (especially age 1–3). Lowest abundances were found in the French NE subpopulation, where the FRA YFS takes place. Nevertheless, trends over all subpopulations were similar with only minor differences in most recent years. Considering the UK BTS is concentrated in the coastal zones and in quarter 3, further investigation is necessary considering the dynamics of these subpopulations and their impact on the overall stock (ICES, 2021).



17.3 Analyses of stock trends/Assessment

17.3.1 Review of last year's assessment

Compared to last year's assessment, the input data (InterCatch data) was updated for the years 2021 to 2023. To assess the impact of these changes, last year's assessment was rerun during the WGNSSK 2025 meeting and the changes were perceived to be ignorable (Figure 17.22).

17.3.2 Final assessment

The SAM model input and configuration used in the WGNSSK 2025 assessment are shown in the table below and Figure 17.23 and Table 17.13. The configuration is the same as the revised SAM model approved during the IBP in 2022.

Settings	
Model	SAM
First data year	1982
Last data year	2024
Ages	1–11+
Plus group	Yes
Stock weights-at-age	Q1 catch weight-at-age; reconstructed for 1982-2003
Discards Numbers- and weight-at-age	Reconstructed for 1982-2003
Abundance indices	Commercial: BEL CBT LPUE (2004-present); FRA COTB LPUE (2005-present); UK CBT LPUE (1986-present)

Settings	
	Survey: UK (E&W) BTS (1989-present; excl. 2022) from 2010-present including only ages 4-6; UK YFS (1987-2006); FRA YFS (1987-2023)
Natural mortality	0.1
Maturity ogive	Age1 = 0.00; Age2 = 0.53; Age3 = 0.92; Age4 = 0.96; Age5 = 0.97; Age6-11+ = 1.00
Number of parameters describing F-at-age in catch (keyLogFsta) (columns represent ages)	0 1 2 3 4 5 5 6 6 7 7 (catch)
Correlation of F across ages (corFlag)	0 (independent)
Number of parameters describing F-at-age in surveys (keyLogFpar) (columns represent ages)	0 (BEL CBT LPUE; FSB) 1 (UK CBT LPUE; FSB) 2 (FRA COTB LPUE; FSB) 3 4 5 6 7 7 (UK BTS; age 1 -6) 8 (UK YFS; age 1) 9 (FRA YFS; age 1)
Density dependent catchability power parameters (keyQpow)	None
Coupling of process variance parameters for F (keyVarF)	0 0 0 0 0 0 0 0 0 0 0
Coupling of process variance parameters for log(N) (keyVarLogN)	0 1 1 1 1 1 1 1 1 1 1
Coupling of variance parameters on the observations (keyVarObs) (columns represent ages)	0 1 2 2 2 2 2 2 2 2 2 (catch; age 1 – 11+) 3 (BEL CBT LPUE; FSB) 4 (UK CBT LPUE; FSB) 5 (FRA COTB LPUE; FSB) 6 7 8 8 8 8 (UK BTS; age 1 - 6) 9 (UK YFS; age 1) 10 (FRA YFS; age 1)
Covariance structure per fleet (obsCorStruct) (columns represent fleets: catch, BEL CBT LPUE, UK CBT LPUE, FRA COTB LPUE, UK BTS, UK YFS, FRA YFS) ID = independent AR = autocorrelated	ID ID ID ID AR ID ID
Coupling of correlation parameters (keyCorObs) (columns represent ages)	0 1 1 1 1 (UK BTS; age 1/2 - age 5/6)
Stock recruitment code (stockRecruitmentModelCode)	0 (random walk)
Number of years where catch scaling is applied (noScaledYears)	None

Settings	
Vector of years where catch scaling is applied (keyScaledYears)	None
Matrix specifying coupling of scale parameters (keyParScaledYA)	None
Fbar ranges	3-7
Type of biomass index (keyBiomassTreat)	2 (fishable stock biomass, FSB)
Option for observational likelihood (obsLikelihoodFlag)	LN LN LN LN LN LN LN
Treatment for weight attribute (fixVarToWeight)	/
Fraction of t(3) distribution used in log(F) increment distribution	/
Fraction of t(3) distribution used in log(N) increment distribution	/
Vector describing fraction for fleets (fracMixObs)	/
Vector describing break year between recruitment (constRecBreaks)	/
Coupling of parameters used in prediction-variance link for observations (predVarObsLink)	None

The SAM model fitting diagnostics and survey catchabilities are shown in Table 17.14 and Table 17.15 respectively. The assessment summary is given in Table 17.16 and Figure 17.24.

Catches are approximately 2000 tonnes lower compared to the beginning of the time-series. The model estimates that the *SSB* ranges between 10 000 and 19 000 tonnes in the period 1982–2024. In 2022, the SSB was estimated at the lowest level of the time-series, but shows a slight increase in 2023 and 2024. Ignoring the plus group, SSB at age is highest for age 6 in 2024 (Table 17.19). As a result of the decline in *fishing pressure* and the low recruitment in recent years, the relative biomass proportion of the plus group increased. However, since 2021, the biomass proportion of the plus group seems to have stabilized to about 14% of the total stock biomass (Figure 17.25).

The *fishing pressure* (F_{bar}) fluctuated over time with periods of high F in the late 1980s, 1990s and 2000s. Since 2010, F_{bar} has gradually declined, from ~0.40, to ~0.23 in 2019. From 2019 to 2022, F_{bar} fluctuated around 0.23, and is estimated to have declined in 2023 and 2024 to a level of 0.215 and 0.207, respectively. The fishing pressure-at-age shows that the age 1 group is hardly caught by the fishery (Figure 17.26; Table 17.18), which is in contrast with all other age groups. The F-at-age shows that the selectivity of the fishery changed remarkably over time. Before 2005, fishing pressure was always highest for age groups 3, 4 and 5, while in the most recent years, fishing pressure for these ages declined strongly. Furthermore, the fishing pressure for age groups 6-11+ also declined with F on ages 10 and 11+ decreasing below F on age 2. In 2024, F-at-age declined for all age groups, except for ages 1 to 3.

The *recruitment* (age 1) is estimated to range between 10 000 and 50 000 thousand individuals, and shows some periods of high recruitment e.g. in the early 1990s and in the period 2000-2010. Since the large recruitment in 2011, recruitment has been stable at lower levels. In 2022, recruitment was estimated at the lowest level of the entire time-series (Table 17.17).

The process residuals do not indicate any major problems with respect to the model configuration (Figure 17.27). However, since 2018, there seems to be a consistent underestimation of the log F processes for the older ages (6-11+).

The one step ahead residuals for the catch data show a consistent underestimation of ages 3-11 in the most recent years (2023 and 2024) (Figure 17.28). In contrast, the UK-BTS indicates mainly positive residuals for ages 3 to 6 in the same period. The UK CBT index residuals shows some clear patterns over the years, which could be explained by the different trend this index shows compared to the other commercial tuning fleets. The French COTB index shows a substantial underestimation in 2023 and 2024.

The retrospective analysis does not indicate large problems with the model with respect to the SSB, Fbar and recruitment estimates (Figure 17.29). All peels fall within the confidence bounds and Mohn's Rho values for SSB and F_{bar} (rho SSB = 0.027; rho F_{bar}(3-7) = -0.018) are within limits as defined at WKFORBIAS (ICES, 2020). The Mohn's Rho for the recruitment (Age 1) is considerably higher (0.233) which is likely related to the lack of informative data sources on age 1.

The leave-one-out runs indicate no strong dependency for one of the tuning fleets (Figure 17.30). However, removing the UK BTS gives a slightly lower SSB and higher F_{bar}, but all within the levels of confidence.

Figure 17.31 gives the model summary compared to the 2023 assessment approved by WGNSSK 2024. There are no clear differences in terms of estimated catches, SSB, F_{bar} and recruitment.

17.3.3 Historical stock trends

Trends in catch, SSB, F_{bar} and recruitment are presented in Table 17.16 and Figure 17.32.

Catches have been fluctuating around 4000 tonnes up to the year 2000. Catches fluctuating around 5000 tonnes were registered for the period 2000–2014. From 2015 onwards, catches dropped below 4000 tonnes and dropped further below 3000 tonnes in 2016 and below 2000 tonnes in 2020 (1971 tonnes). In 2023, the lowest catches of the time-series have been realised 1504 tonnes). A slight increase was observed in 2024 (1583 tonnes).

The *spawning-stock biomass* (SSB) has been fluctuating without trend since the 1980s between MSY $B_{trigger}$ and B_{lim}. In the period 2000–2015, SSB has been fluctuating around MSY $B_{trigger}$. However, since 2019, SSB has been fluctuating around B_{lim} and is assessed to be below B_{lim} in 2024.

Fishing pressure (F) has been exceeding F_{lim} (0.352) for the major part of the time-series. From 2013 onwards, F decreased to below F_{pa} (0.318), decreasing further to drop below F_{MSY} (0.230) in 2020. Since then, fishing pressure stabilized at levels just below F_{MSY}.

Recruitment has been fluctuating without trend with occasional strong year classes. The last 10 years, recruitment has been lower compared to the earlier part of the time-series.

17.4 Short-term forecast

Since the last benchmark (WKNSEA 2021), the short term forecast of sole in Division 27.7d is performed using the *stockassessment* package. Stock weights-at-age for the next three years are assumed to be the mean stock weight-at-age of the last five years. Selectivity of the fishery for the next three years is assumed to be the mean selectivity of the last three years. As the selectivity of the fishery is considerably changing over time, it was assumed that a 3 year average would be more appropriate than a 5 year average as was done last year.

Recruitment in the future years is resampled from 2012 until the final assessment year model estimates (2012–2024). To account for the higher uncertainty of the final recruitment estimates, a weighted resampling procedure is applied. Hereto, resampling probabilities are defined as the inverse of the standard deviation of the recruitment estimates. This was done to be more consistent with the forecast procedure implemented in the *stockassessment* package that also considers the last recruitment estimate as a starting point for the projection. A stochastic forecast is conducted with 1000 simulations. The process noise F is set to false. The forecast results are split into landings and discards with the proportion of landings calculated as the mean of the last 5 years. An overview of the forecast settings is provided in the stock annex.

For the fishing pressure in the intermediate year, there are two possible scenarios: 1) status quo fishing pressure (F_{sq}) or 2) TAC constraint. For the status quo fishing pressure, there are again two options: 1a) if the F_{bar} shows no trend over the last three years, the mean F_{bar} of the last three years is taken as intermediate year assumption, 1b) if the F_{bar} shows a decreasing or increasing trend over the last three years, we scale to the last data year, which means that the F_{bar} in the intermediate years is the same as the last data year. For the TAC constraint option, the F_{bar} is calculated in the intermediate year as if the TAC would be fully fished in that year.

For this year's assessment, the status quo F (trend, F of 2024) was considered given the rather stable pattern in F and the decline in the TAC for the year 2025. Given the landing obligation exemptions for sole, the strong decline in TAC, and the fact that the main fisheries fished their entire quotum in 2024, it was assumed that the TAC in 2025 counted against the landings, which corresponds in this case to F status quo is more likely than applying a TAC constraint to the catches in 2025.

SSB 2026	**F_{bar} (age 3–7)**	**F_{dis}**	**F_{lan}**	**recruits (age 1; thousands)**
11 009 t	0.207	0.045	0.162	14 696
Landings	**Discards**	**Catch**	**TAC 2025**	**Catch advice for 2025**
1209 t	326 t	1535 t	1209 t	1209 t

Following the ICES advice rules, the target F in the advice year is set at F_{MSY} in case the SSB in the advice year (2026) is above MSY $B_{trigger}$, else, the target F is set as F_{MSY} x ($SSB_{advice\_year}/B_{trigger}$). In case the SSB is insufficient to bring the stock above B_{lim} in the advice year + 1, a zero TAC can be advised.

The SSB in 2026 (11009 tonnes) is below MSY $B_{trigger}$ (15654 tonnes), which means that F_{MSY} should be rescaled with a factor of 0.707 (MSY approach) to 0.162. This resulted in a catch advice for 2026 of 1275 tonnes, which is a 5.5% increase compared to the catch advice for 2025 (1209 tonnes). This means a probability of SSB 2027 to fall below B_{lim} of 35%.

The change in advice (+5.5%) is the result of the higher recruitment estimated for 2024 than assumed last year, and higher target F for the advice year. The evolution of stock weights is shown in Figure 17.33. Comparisons between selectivity, stock weights, stock numbers, and change in biomass at age between this year's estimates and last year's forecast assumptions are given in Figures 17.34 to 17.39.

The output of the forecast, for the F_{sq} option in the intermediate year is shown in the table below. MSY $B_{trigger}$ and B_{pa} cannot be reached in 2027, therefore, the scenario cannot be calculated.

Sole in Division 7.d. Annual catch scenarios. All weights are in tonnes.

Basis	Total catch (2026)	Projected landings (2026)	Projected discards^ (2026)	F_{total} (ages 3–7) (2026)	$F_{projected\ landings}$ (ages 3–7) (2026)	$F_{projected\ discards}$ (ages 3–7) (2026)	Spawning-stock biomass (SSB; 2027)	% SSB change*	% total allowable catch (TAC) change**	% advice change**	Probability of SSB (2027) < B_{lim} (%) ***
ICES advice basis											
Maximum sustainable yield (MSY) approach: $F_{MSY} \times$ SSB (2026)/MSY $B_{trigger}$	1275	1002	273	0.162	0.126	0.036	11860	7.7	5.5	5.5	35
Other scenarios											
$F_{MSY\ lower} \times$ SSB (2026)/ MSY $B_{trigger}$	889	699	190	0.110	0.086	0.024	12309	11.8	−26	−26	26
F = 0	0	0	0	0	0	0	13324	21	−100	−100	11.6
F_{PA}	2322	1833	489	0.318	0.25	0.069	10652	−3.2	92	92	62
SSB (2027) = B_{lim}	1873	1475	398	0.248	0.194	0.054	11181	1.56	55	55	50
SSB (2027)= B_{PA} = MSY $B_{trigger}$#											
F = F_{2025}	1596	1256	340	0.207	0.162	0.045	11495	4.4	32	32	44
$F_{MSY\ lower}$	1235	971	264	0.156	0.122	0.034	11904	8.1	2.2	2.2	34
F_{MSY}	1751	1379	372	0.230	0.180	0.050	11315	2.8	45	45	47

^ Including below minimum size (BMS) landings, assuming recent discard rate.

* SSB 2027 relative to SSB 2026.

** Total advised catch in 2026 relative to the advice value 2025 and TAC (both 1209 tonnes).

*** The probability of SSB being below B_{lim} in 2027. This probability relates to the short-term probability of SSB < B_{lim} and is not comparable to the long-term probability of SSB < B_{lim} tested in simulations when estimating fishing mortality reference points.

\# B_{PA} and MSY $B_{trigger}$ cannot be achieved in 2027, even with zero catches.

17.5 Biological reference points

The table below summarizes all known reference points for sole in Division 27.7.d and their technical basis. Reference points have been redefined as a result of the 2022 IBP (ICES, 2022).

Framework	Reference point	Value	Technical basis	Source
MSY approach	MSY $B_{trigger}$	15654	B_{pa}; in tonnes.	ICES (2022)
	F_{MSY}	0.230	Stochastic simulations (EqSim) with segmented regression fixed at B_{lim} based on recruitment period 1982–2020.	ICES (2022)
Precautionary approach	B_{lim}	11181	B_{loss}, lowest observed SSB (2020) with 2021 as the last year of catch data; in tonnes.	ICES (2022)
	B_{pa}	15654	B_{lim} × 1.4; in tonnes.	ICES (2022)
	F_{lim}	0.352	The F that on average leads to B_{lim} from EqSim.	ICES (2022)
	F_{pa}	0.318	The F that provides a 95% probability for SSB to be above B_{lim} ($F_{P.05}$ with AR)	ICES (2022)
F_{MSY} ranges	F_{lower}	0.156–0.230	Consistent with ranges resulting in no more than 5% reduction in long-term yield compared with F_{MSY}	ICES (2016, 2022)
	F_{upper}	0.230–0.287	Consistent with ranges resulting in no more than 5% reduction in long-term yield compared with F_{MSY}	ICES (2016, 2022)

17.6 Quality of the assessment

The recruitment estimates of the years 2023 and 2024 are highly uncertain due to a lack of data that informs the dynamics of the age 1 individuals. These recruitment estimates may be biased with knock-on effects on the short-term forecast and advice. However, the uncertainty is accounted for in the forecasted recruitment by weighting the resampled recruitment with the associated standard deviation.

17.7 Benchmark issue list

17.7.1 Data issues

During the benchmark, it was noted that sole in Division 27.7d exhibited a declining trend in the in the weights and lengths-at-age in recent years and more apparent in the older ages. It is not clear what mechanism is driving such decline. Future work should look into the potential causes for this declining trend, as well as potential effects on the selection pattern of the scientific survey which is assumed to be constant over time.

Maturity estimates were not investigated during the last benchmark (WKNSEA 2021; ICES, 2021). Therefore, maturity estimates as calculated during the WKNSEA 2017 benchmark are used. These are derived from both commercial landings and survey data. Using commercial data could potentially introduce bias. When maturity estimates are revised, only fishery independent data should be considered (if available) to ensure that they align with contemporary stock dynamics. Future work should also revisit growth and natural mortality.

The subpopulation structure in this stock should be investigated further.

To improve estimation of discards in the assessment, discard survival by gear type could be considered.

The poorer tracking of cohorts and internal consistency in the most recent part of the UK BTS time-series should be investigated further.

17.7.2 Assessment issues

Biological and environmental processes should be explored for potential use in a model. There is also an increase of the biomass proportion in the older age groups and notably the plus group that should be followed up in future assessments. However, this trend seems to have stabilised in this year's assessment.

The retrospective pattern for the recruitment is higher as there is little information to inform the dynamics of the younger ages (1-3).

Alternative methods to calculate the reference points should be investigated, especially for this stock, which is characterised by a narrow range in SSB and no clear stock-recruitment relationship.

17.7.3 Short-term forecast issues

Currently no issues.

17.8 Management considerations

The sole stock in Division 27.7.d is harvested in a mixed fishery with plaice in 27.7.d. Due to the minimum mesh size in the mixed beam and otter trawl fisheries (80 mm), a large number of undersized plaice are discarded. The 80 mm mesh size is not matched to the minimum landing size of plaice (27 cm). Measures taken specifically to control sole fisheries will impact the plaice fisheries.

17.9 References

Archambault B., Le Pape O., Baullier L., Vermard Y., Rivot E., 2016. Adults mediated connectivity affects inferences on population dynamics and stock assessment of exploited marine populations. Fisheries Research, 181: 198-213.

EU. 2018. Commission delegated regulation (EU) 2018/2034. Establishing a discard plan for certain demersal fisheries in North-Western waters for the period 2019-2021. Official Journal of the European Union. 327/8.

ICES. 2016. Report of the Workshop to consider FMSY ranges for stocks in ICES categories 1 and 2 in Western Waters (WKMSYREF4), 13–16 October 2015, Brest, France. ICES CM 2015/ACOM:58. 187 pp.

ICES. 2017. Report of the Benchmark Workshop on North Sea Stocks (WKNSEA), 6–10 February 2017, Copenhagen, Denmark. ICES CM 2017/ACOM:34. 673 pp.

ICES. 2019. Inter-benchmark Protocol for sole in the Eastern English Channel (IBPsol7d). ICES Scientific Reports. 1:75. 88 pp. http://doi.org/10.17895/ices.pub.5631

ICES. 2020. Workshop on Catch Forecast from Biased Assessments (WKFORBIAS; outputs from 2019 meeting). ICES Scientific Reports. 2:38. 38pp. http://doi.org/10.17895/ices.pub.5997

ICES. 2021. Benchmark Workshop on North Sea Stocks (WKNSEA). ICES Scientific Reports. 3:25. 768 pp. https://doi.org/10.17895/ices.pub.7922

ICES. 2022. ICES Inter-benchmark protocol on eastern English Channel sole (IBPSOL7d). ICES Scientific Reports. 4:67. 111 pp. http://doi.org/10.17895/ices.pub.20489484

Lecomte, J.-B., Le Pape, O., Baillif, H., Nevoux, M., Vermard, Y., Savina, M., Veron, M., Lehuta, S., Hunter, E., Rivot, E., 2020. State-space modelling of multidecadal mark-recapture data reveals low adult dispersal in a nursery-dependent fish metapopulation. Can. J. Fish. Aquat. Sci., 77: 342-354. Dx.doi.org/10.1139/djfas-2019-0037.

Randon, M., Réveillac, E., Rivot, E., Du Pontavice, H., Le Pape, O., 2018. Could we consider a single stock when spatial sub-units present lasting patterns in growth and asynchrony in cohort densities? A flatfish case study. Journal of Sea Research, 142, 91-100.

Randon, M., Le Pape, O., Ernande, B., Mahé, K., Volckaert, F.A.M., Petit, E.J., Lassale, G., Le Berre, T., Réveillac, E., 2020. Complementarity and discriminatory power of genotype and otolith shape in describing the fine-scale population structure of an exploited fish, the common sole of the Eastern English Channel. Plos One, https://doi.org/10.1371/journal.pone.0241429

17.10 Tables and figures

Table 17.1. Sole in Division 7.d. Official landings (tonnes) by country over the period 1974–2024; ICES estimates (as reported in InterCatch) for both landings and discards (tonnes) used by the working group. TAC (tonnes) represents landings until 2015. From 2016 onwards, TAC represents catch. * including BMS.

Year	Official Landings					ICES estimates		TAC
	Belgium	France	UK (E&W)	Other	Total	Landings	Discards	
1974	159	383	309	3	854	884		
1975	132	464	244	1	841	882		
1976	203	599	404		1206	1305		
1977	225	737	315		1277	1335		
1978	241	782	366		1389	1589		
1979	311	1129	402		1842	2215		
1980	302	1075	159		1536	1923		
1981	464	1513	160		2137	2477		
1982	525	1828	317	4	2674	3190	196	
1983	502	1120	419		2041	3458	101	
1984	592	1309	505		2406	3575	141	
1985	568	2545	520		3633	3837	242	
1986	858	1528	551		2937	3932	145	
1987	1100	2086	655		3841	4791	197	3850

Year	Official Landings					ICES estimates		TAC
	Belgium	France	UK (E&W)	Other	Total	Landings	Discards	
1988	667	2057	578		3302	3853	198	3850
1989	646	1610	689		2945	3805	192	3850
1990	996	1255	785		3036	3647	342	3850
1991	904	2054	826		3784	4351	368	3850
1992	891	2187	706	10	3794	4072	275	3500
1993	917	2322	610	13	3862	4299	273	3200
1994	940	2382	701	15	4038	4383	122	3800
1995	817	2248	669	9	3743	4420	282	3800
1996	899	2322	877		4098	4797	174	3500
1997	1306	1702	933		3941	4764	147	5230
1998	541	1703	803		3047	3363	127	5230
1999	880	2251	769		3900	4135	247	4700
2000	1021	2190	621		3832	3476	201	4100
2001	1313	2482	822		4617	4025	317	4600
2002	1643	2780	976		5399	4733	444	5200
2003	1657	3475	1114	1	6247	6977	584	5400
2004	1485	3070	1112		5667	5819	258	5900
2005	1221	2832	567		4620	4748	344	5700
2006	1547	2627	658	0	4832	4830	315	5720
2007	1530	2981	801	1	5313	5421	332	6220
2008	1368	2880	724	0	4972	4963	183	6590
2009	1475	3047	760	0	5282	4828	287	5274
2010	1294	2476	679	0	4449	4108	273	4219
2011	1222	2281	700	0	4203	4136	342	4852
2012	941	2475	627	0	4043	4058	445	5580
2013	952	2884	605	0	4441	4295	180	5900
2014	1496	2507	648	< 1	4651	4626	216	4838
2015	1048	1895	468	0	3411	3385	263	3483

Year	Official Landings					ICES estimates		TAC
	Belgium	France	UK (E&W)	Other	Total	Landings	Discards	
2016	799	1337	392	< 1	2528	2433	106	3258
2017	697*	1178	349	< 1	2224	2090	156	2724
2018	653	1265	394	< 1	2312	2395	263	3405
2019	603*	925	245*	< 1	1773	1648	404	2515
2020	686*	827	199*	< 1	1712	1562	409*	2797
2021	623*	824	233*	1	1681	1567	348*	3248
2022	644*	884*	262	1	1791	1699	316	2380
2023	538*	567	184	< 1	1289	1293	211	1747
2024	412*	519*	136	< 1	1067	1112	471	1504
2025								1209

Table 17.2. Sole in Division 7.d. Summary of the InterCatch data in 2024 (imported vs. raised data; sampled vs. estimated data).

CatchCategory	RaisedOrImported	SampledOrEstimated	CATON	perc
Landings	Imported_Data	Sampled_Distribution	953.2	86
Landings	Imported_Data	Estimated_Distribution	158.6	14
Discards	Imported_Data	Sampled_Distribution	343.8	73
Discards	Imported_Data	Estimated_Distribution	0.298	0
Discards	Raised_Discards	Estimated_Distribution	127	27
BMS landing	Imported_Data	Estimated_Distribution	0	NA

Table 17.3. Sole in Division 7.d. Landings percentages by gear type for 2017–2024 (GNS/GTR = gill and trammel nets; TBB = beam trawls; OTB/OTT/SSC/SDN = otter trawls and seines; other gears include MIS/FPO/DRB/LHM/LLS).

Landings by gear	2017	2018	2019	2020	2021	2022	2023	2024
GNS/GTR	46%	44%	33%	28%	32%	31%	29%	30%
TBB	31%	28%	33%	36%	35%	34%	45%	43%
OTB/OTT/SSC/SDN	17.6%	24%	30%	33%	30%	30%	24%	21%
Other	4.7%	4.5%	3.7%	2.8%	2.7%	3.9%	2.6%	7%

Table 17.4. Sole in Division 7.d. Catch numbers at age (in thousands).

age	1982	1983	1984	1985	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995
1	375	0	59	122	122	22	236	405	3092	952	261	211	77	2082
2	3432	1136	2630	4961	1685	4197	2910	4686	3836	9646	5446	6769	934	4006
3	5688	3812	3476	5795	5904	4158	7995	3586	6214	4575	9794	7179	6912	4874
4	1710	3971	2630	1675	3259	3336	1633	4482	1172	4242	1925	5551	6017	4857
5	558	895	1890	1032	911	2068	1167	1443	1505	608	2006	1015	3427	2987
6	636	731	736	1863	771	1046	859	842	302	1000	289	565	586	1986
7	535	624	454	145	1062	1095	390	574	392	258	370	163	570	377
8	233	330	313	158	155	785	255	201	260	247	135	188	109	278
9	118	107	134	156	190	111	256	166	129	258	171	116	147	88
10	81	88	98	69	212	163	83	224	126	92	95	62	93	106
11+	196	191	235	128	372	459	275	282	489	382	231	129	258	241

age	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009
1	22	60	82	417	343	418	1756	57	121	771	412	168	826	1270
2	2456	2004	1855	4395	4831	8136	9431	15482	4164	6957	6942	7620	2872	4446
3	8650	6761	6259	9470	5412	6905	8367	10669	11013	5185	4285	8307	8562	4494
4	3094	5106	2761	3369	4485	1627	3839	4069	3682	4777	3097	3169	5679	6164
5	3227	2096	1649	1319	1084	2509	1422	2168	4595	1256	3316	1794	1452	2500
6	1830	1676	612	871	507	731	657	656	1670	920	1207	2769	1086	808
7	1289	920	562	352	320	291	299	2068	379	636	1128	1010	758	719
8	271	776	443	672	148	128	129	229	394	392	579	753	410	664
9	319	239	354	351	328	56	97	73	291	211	239	450	157	277
10	112	169	239	192	150	81	57	134	254	104	233	194	168	239
11+	344	267	301	359	248	265	197	285	443	266	383	473	276	425

age	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
1	585	353	739	40	372	300	144	565	1643	664	7	1040	433	545	1176
2	5827	6148	3759	1150	1244	2131	1145	1060	3378	2341	2736	1119	2164	1314	3431
3	4255	6938	8544	5951	3502	2101	2185	2467	1846	3086	2683	3426	1490	2020	1419

age	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
4	2953	2854	5253	6595	6639	2303	1253	1447	2626	1086	2223	1705	2512	801	1133
5	3034	1562	1433	2539	4259	3496	1308	826	1022	1476	812	1225	1461	1375	431
6	1621	1469	930	762	1853	2555	1553	876	736	481	915	471	844	447	650
7	320	562	563	545	687	1194	1059	850	619	312	427	396	441	400	239
8	277	178	414	535	417	463	598	698	821	227	166	244	385	105	183
9	288	147	98	205	374	142	188	287	451	392	131	110	197	163	49
10	102	132	46	59	145	238	211	139	286	282	182	75	74	74	79
11+	376	179	259	129	255	272	322	194	292	230	331	224	221	118	115

Table 17.5. Sole in Division 7.d. Catch weights at age (kg)

age	1982	1983	1984	1985	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995
1	0.078	0.000	0.076	0.068	0.103	0.073	0.078	0.081	0.091	0.087	0.078	0.065	0.076	0.099
2	0.155	0.157	0.162	0.166	0.164	0.159	0.138	0.140	0.162	0.146	0.139	0.134	0.136	0.160
3	0.215	0.220	0.224	0.220	0.203	0.226	0.216	0.184	0.228	0.199	0.194	0.189	0.178	0.171
4	0.309	0.299	0.311	0.279	0.303	0.292	0.276	0.269	0.287	0.264	0.265	0.245	0.233	0.228
5	0.385	0.403	0.379	0.367	0.362	0.352	0.359	0.292	0.348	0.353	0.289	0.334	0.287	0.254
6	0.427	0.435	0.435	0.393	0.386	0.406	0.408	0.357	0.339	0.393	0.402	0.383	0.354	0.332
7	0.439	0.434	0.416	0.515	0.436	0.410	0.458	0.387	0.469	0.421	0.390	0.536	0.380	0.356
8	0.509	0.524	0.538	0.543	0.520	0.482	0.514	0.472	0.465	0.430	0.462	0.553	0.505	0.385
9	0.502	0.537	0.529	0.594	0.502	0.465	0.553	0.515	0.487	0.434	0.459	0.515	0.484	0.490
10	0.463	0.583	0.565	0.595	0.523	0.538	0.563	0.547	0.518	0.478	0.463	0.766	0.496	0.494
11+	0.672	0.628	0.714	0.800	0.602	0.618	0.665	0.701	0.562	0.566	0.566	0.667	0.616	0.654

age	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009
1	0.109	0.106	0.101	0.099	0.110	0.082	0.091	0.101	0.097	0.123	0.133	0.095	0.121	0.113
2	0.150	0.139	0.145	0.137	0.129	0.138	0.147	0.148	0.147	0.158	0.150	0.158	0.156	0.155
3	0.170	0.180	0.165	0.181	0.169	0.202	0.195	0.218	0.181	0.191	0.192	0.175	0.207	0.201
4	0.227	0.231	0.233	0.213	0.221	0.281	0.251	0.286	0.238	0.262	0.233	0.201	0.243	0.252
5	0.268	0.291	0.285	0.259	0.331	0.287	0.315	0.365	0.269	0.353	0.286	0.267	0.258	0.268
6	0.323	0.342	0.343	0.280	0.376	0.333	0.375	0.407	0.293	0.434	0.338	0.280	0.311	0.322

age	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009
7	0.360	0.389	0.382	0.290	0.424	0.367	0.376	0.165	0.410	0.455	0.394	0.339	0.370	0.316
8	0.405	0.404	0.417	0.341	0.427	0.374	0.393	0.474	0.449	0.490	0.425	0.387	0.397	0.383
9	0.435	0.503	0.484	0.358	0.384	0.493	0.469	0.424	0.390	0.566	0.562	0.452	0.433	0.383
10	0.465	0.474	0.435	0.374	0.459	0.511	0.420	0.504	0.487	0.648	0.497	0.424	0.511	0.430
11+	0.585	0.651	0.616	0.535	0.680	0.544	0.531	0.565	0.664	0.550	0.552	0.570	0.509	0.484

age	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
1	0.101	0.089	0.058	0.095	0.098	0.077	0.093	0.094	0.081	0.058	0.091	0.083	0.094	0.133	0.099
2	0.152	0.148	0.121	0.141	0.134	0.127	0.136	0.143	0.121	0.118	0.114	0.124	0.132	0.129	0.131
3	0.192	0.201	0.181	0.189	0.176	0.168	0.199	0.184	0.171	0.175	0.152	0.164	0.153	0.174	0.171
4	0.241	0.245	0.233	0.241	0.231	0.223	0.242	0.229	0.214	0.230	0.197	0.195	0.193	0.217	0.225
5	0.276	0.301	0.270	0.297	0.267	0.266	0.266	0.252	0.268	0.244	0.224	0.240	0.233	0.223	0.253
6	0.322	0.330	0.312	0.301	0.325	0.282	0.285	0.291	0.289	0.274	0.243	0.277	0.265	0.280	0.272
7	0.334	0.357	0.375	0.384	0.328	0.330	0.320	0.293	0.289	0.290	0.271	0.291	0.255	0.298	0.293
8	0.337	0.424	0.354	0.402	0.389	0.329	0.371	0.362	0.250	0.318	0.312	0.320	0.307	0.356	0.325
9	0.367	0.389	0.424	0.415	0.413	0.408	0.361	0.432	0.327	0.272	0.382	0.373	0.332	0.353	0.397
10	0.520	0.425	0.544	0.463	0.494	0.372	0.358	0.479	0.362	0.338	0.285	0.409	0.372	0.382	0.370
11+	0.502	0.534	0.521	0.572	0.527	0.480	0.436	0.525	0.409	0.394	0.417	0.465	0.402	0.491	0.415

Table 17.6. Sole in Division 7.d. Stock weights at age (kg)

age	1982	1983	1984	1985	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996
1	0.077	0.078	0.075	0.067	0.102	0.072	0.077	0.080	0.090	0.086	0.077	0.064	0.075	0.098	0.108
2	0.167	0.169	0.175	0.179	0.177	0.172	0.149	0.151	0.175	0.157	0.150	0.145	0.147	0.173	0.162
3	0.243	0.248	0.253	0.248	0.229	0.255	0.244	0.208	0.257	0.225	0.219	0.213	0.201	0.193	0.192
4	0.352	0.340	0.354	0.318	0.345	0.332	0.314	0.306	0.327	0.301	0.302	0.279	0.265	0.260	0.258
5	0.424	0.444	0.417	0.404	0.399	0.388	0.395	0.322	0.383	0.389	0.318	0.368	0.316	0.280	0.295
6	0.472	0.481	0.481	0.435	0.427	0.449	0.451	0.395	0.375	0.435	0.445	0.424	0.392	0.367	0.357
7	0.488	0.482	0.462	0.572	0.485	0.456	0.509	0.430	0.521	0.468	0.434	0.596	0.422	0.396	0.400
8	0.441	0.454	0.466	0.471	0.451	0.418	0.445	0.409	0.403	0.373	0.400	0.479	0.438	0.334	0.351
9	0.486	0.520	0.512	0.575	0.486	0.450	0.536	0.499	0.472	0.420	0.445	0.499	0.469	0.475	0.421

age	1982	1983	1984	1985	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996
10	0.481	0.606	0.587	0.618	0.543	0.559	0.585	0.568	0.538	0.497	0.481	0.796	0.515	0.513	0.483
11+	0.720	0.676	0.766	0.861	0.646	0.662	0.713	0.754	0.604	0.607	0.606	0.718	0.662	0.701	0.629

age	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011
1	0.105	0.100	0.098	0.109	0.081	0.090	0.100	0.097	0.123	0.136	0.087	0.115	0.113	0.101	0.089
2	0.150	0.156	0.148	0.139	0.149	0.159	0.160	0.142	0.154	0.142	0.140	0.141	0.148	0.129	0.119
3	0.203	0.186	0.204	0.191	0.228	0.220	0.246	0.179	0.189	0.182	0.168	0.190	0.196	0.193	0.188
4	0.263	0.265	0.242	0.252	0.320	0.286	0.326	0.225	0.262	0.241	0.220	0.246	0.264	0.260	0.250
5	0.320	0.314	0.285	0.365	0.316	0.347	0.402	0.265	0.361	0.316	0.282	0.249	0.288	0.292	0.313
6	0.378	0.379	0.310	0.416	0.368	0.415	0.450	0.285	0.443	0.352	0.315	0.338	0.344	0.363	0.338
7	0.432	0.425	0.322	0.471	0.408	0.418	0.183	0.409	0.466	0.398	0.346	0.333	0.304	0.363	0.371
8	0.350	0.361	0.295	0.370	0.324	0.341	0.411	0.510	0.514	0.465	0.443	0.435	0.438	0.371	0.481
9	0.487	0.469	0.347	0.372	0.477	0.454	0.411	0.391	0.598	0.574	0.496	0.373	0.352	0.421	0.409
10	0.493	0.452	0.389	0.477	0.531	0.436	0.524	0.514	0.704	0.496	0.397	0.713	0.437	0.542	0.458
11+	0.700	0.661	0.577	0.727	0.585	0.569	0.608	0.692	0.588	0.632	0.613	0.472	0.606	0.537	0.561

Age	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
1	0.057	0.092	0.097	0.075	0.090	0.094	0.081	0.060	0.093	0.082	0.076	0.098	0.098
2	0.104	0.104	0.093	0.102	0.101	0.135	0.111	0.104	0.097	0.101	0.091	0.106	0.125
3	0.177	0.163	0.144	0.177	0.211	0.176	0.177	0.177	0.142	0.173	0.129	0.166	0.167
4	0.228	0.244	0.235	0.244	0.273	0.229	0.247	0.232	0.203	0.220	0.194	0.225	0.224
5	0.275	0.339	0.283	0.296	0.294	0.267	0.296	0.272	0.263	0.274	0.233	0.237	0.257
6	0.331	0.340	0.346	0.308	0.331	0.305	0.324	0.305	0.273	0.311	0.268	0.301	0.306
7	0.387	0.439	0.396	0.373	0.367	0.323	0.343	0.307	0.317	0.331	0.272	0.329	0.374
8	0.384	0.416	0.429	0.336	0.448	0.384	0.332	0.352	0.369	0.371	0.306	0.391	0.368
9	0.467	0.431	0.442	0.398	0.537	0.478	0.371	0.286	0.459	0.398	0.347	0.382	0.502
10	0.548	0.416	0.592	0.380	0.456	0.508	0.407	0.361	0.346	0.416	0.352	0.374	0.470
11+	0.573	0.683	0.541	0.519	0.580	0.575	0.463	0.457	0.495	0.537	0.454	0.557	0.489

Table 17.7. Sole in Division 7.d. Tuning series 1: revised Belgian commercial beam trawl LPUE (2004–2024).

	Effort	Biomass index
2004	1	14.72204251
2005	1	14.95745776
2006	1	16.02364983
2007	1	16.71950689
2008	1	15.28191325
2009	1	19.06356295
2010	1	19.41034228
2011	1	17.32139264
2012	1	18.43170695
2013	1	18.52714202
2014	1	24.89342401
2015	1	18.19680586
2016	1	14.45257677
2017	1	12.01916843
2018	1	12.78568645
2019	1	11.70561923
2020	1	13.37807871
2021	1	12.89287361
2022	1	12.93832509
2023	1	12.13986061
2024	1	11.19071511

Table 17.8. Sole in Division 7.d. Tuning series 2: revised French commercial otter trawl LPUE (2005–2024).

	Effort	Biomass index
2005	1	13.27654273
2006	1	18.98994703
2007	1	16.07989295
2008	1	19.60347081
2009	1	17.86384112

	Effort	Biomass index
2010	1	22.64104298
2011	1	22.29687739
2012	1	18.34728866
2013	1	20.42425728
2014	1	22.77504759
2015	1	15.1899479
2016	1	15.85655742
2017	1	13.68850757
2018	1	19.10079272
2019	1	15.27052429
2020	1	13.89853034
2021	1	13.61437138
2022	1	12.34374796
2023	1	8.166289604
2024	1	5.259793771

Table 17.9. Sole in Division 7.d. Tuning series 3: UK (E&W) commercial beam trawl LPUE (1986–2024).

	Effort	Biomass index
1986	1	138.72492
1987	1	142.8913114
1988	1	133.2336623
1989	1	107.7585952
1990	1	109.8050754
1991	1	71.17088069
1992	1	65.53329156
1993	1	53.93743589
1994	1	55.80003059
1995	1	65.99439722
1996	1	91.56366808

	Effort	Biomass index
1997	1	88.65710825
1998	1	101.4844797
1999	1	94.33305591
2000	1	93.5978889
2001	1	98.05895762
2002	1	128.4165414
2003	1	118.3101589
2004	1	125.8325338
2005	1	141.4056712
2006	1	125.3070046
2007	1	120.9682142
2008	1	108.3932023
2009	1	85.26167953
2010	1	86.47515299
2011	1	82.65608545
2012	1	77.33684063
2013	1	81.5939132
2014	1	87.09997481
2015	1	92.38899153
2016	1	97.06890425
2017	1	76.36326451
2018	1	77.31401802
2019	1	73.77754664
2020	1	86.43207608
2021	1	94.30956124
2022	1	106.8612657
2023	1	100.5415222
2024	1	84.16520822

Table 17.10. Sole in Division 7.d. Tuning series 4: UK (E&W) beam trawl survey (Q3) (1989–2024); grey values are currently not included in the assessment.

	Effort	Age1	Age2	Age3	Age4	Age5	Age6
1989	1	3.01	22.09	4.62	2.45	0.56	0.35
1990	1	17.96	5.55	5.55	1.24	1.01	0.33
1991	1	12.14	31.17	3.19	2.82	0.48	0.67
1992	1	1.33	15.29	13.47	1.07	1.61	0.34
1993	1	0.82	22.96	11.42	9.97	1.14	1.52
1994	1	8.33	4.26	11.07	4.65	4.3	0.28
1995	1	5.89	16.09	2.22	3.51	1.67	2.12
1996	1	5.3	10.79	5.97	1.07	1.86	1.15
1997	1	24.75	10.85	4.42	1.94	0.26	0.82
1998	1	3.27	24.11	3.67	1.47	0.83	0.19
1999	1	35.99	8.22	11.33	1.59	0.73	1.02
2000	1	14.98	27.45	5.52	4.85	1.48	0.68
2001	1	10.19	27.88	11.55	1.67	2.33	0.75
2002	1	53.56	16.11	8.6	5.11	0.45	1.04
2003	1	11.03	45.65	5.87	3.2	2.05	0.42
2004	1	12.67	11.81	10.97	2.08	2.02	1.34
2005	1	43.27	6.91	3.5	5.18	1.9	1.15
2006	1	10.84	42.62	4.51	2.68	2.59	0.55
2007	1	2.57	28.97	15.45	1.47	1.04	1.56
2008	1	3.77	7.35	9.14	5.82	0.4	0.68
2009	1	51.25	19.16	7.1	5.81	5.02	0.44
2010	1	16.59	30.76	5.14	1.66	2.7	2.73
2011	1	13.66	28.6	14.7	1.66	0.54	2.62
2012	1	1.75	9.72	7.51	3.53	0.92	0.39
2013	1	0.72	8.91	15.09	9.72	3.23	1.12
2014	1	25.39	16.35	12.38	11.92	5.09	2.73
2015	1	25.24	21.36	6.04	2.29	4.51	2.08
2016	1	10.17	33.14	11.17	3.16	3.17	3.02

	Effort	Age1	Age2	Age3	Age4	Age5	Age6
2017	1	27.85	15.18	16.26	2.67	2.13	1.52
2018	1	14.86	36.49	6.66	10.32	1.74	2.13
2019	1	56.54	31.08	19.53	1.18	4.01	2.53
2020	1	1.87	42.73	8.01	4.62	1.15	1.84
2021	1	32.95	16.06	22.82	5.31	4.65	1.10
2022	/	/	/	/	/	/	/
2023	1	19.55	15.49	10.26	2.91	6.26	0.84
2024	1	5.52	10.54	4.00	2.36	1.06	2.48

Table 17.11. Sole in Division 7.d. Tuning series 5: UK (E&W) young fish survey (1987–2006).

	Effort	Age1
1987	1	1.38
1988	1	1.87
1989	1	0.62
1990	1	1.9
1991	1	3.69
1992	1	1.5
1993	1	1.33
1994	1	2.68
1995	1	2.91
1996	1	0.57
1997	1	1.12
1998	1	1.12
1999	1	1.47
2000	1	2.47
2001	1	0.38
2002	1	4.15
2003	1	1.44
2004	1	2.72
2005	1	4.07

	Effort	Age1
2006	1	2.21

Table 17.12. Sole in Division 7.d. Tuning series 6: French young fish survey (1987–2024) funded by EDF (noursom).

	Effort	Age1
1987	1	0.07
1988	1	0.17
1989	1	0.14
1990	1	0.54
1991	1	0.38
1992	1	0.22
1993	1	0.03
1994	1	0.7
1995	1	0.28
1996	1	0.15
1997	1	0.03
1998	1	0.1
1999	1	0.35
2000	1	0.31
2001	1	1.21
2002	1	0.11
2003	1	0.32
2004	1	0.15
2005	1	0.82
2006	1	0.83
2007	1	0.08
2008	1	0.06
2009	1	2.78
2010	1	0.1
2011	1	0.32

	Effort	Age1
2012	1	0.35
2013	1	0.052
2014	1	0.04
2015	1	0.09
2016	1	0.04
2017	1	0.05
2018	1	0.03
2019	1	0.45
2020	1	0.38
2021	1	0.07
2022	1	0.01
2023	1	0.14
2024	1	No data point available

Table 17.13. Sole in Division 7.d. SAM model configuration of the 2025 assessment

```
# Where a matrix is specified rows corresponds to fleets and columns to ages.
# Same number indicates same parameter used
# Numbers (integers) starts from zero and must be consecutive

$minAge
# The minimum age class in the assessment
 1
$maxAge
# The maximum age class in the assessment
 11
$maxAgePlusGroup
# Is last age group considered a plus group (1 yes, or 0 no).
 1 0 0 0 0 0 0

$keyLogFsta
# Coupling of the fishing mortality states (normally only first row is used).
     [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10] [,11]
[1,]    0    1    2    3    4    5    5    6    6     7     7
[2,]   -1   -1   -1   -1   -1   -1   -1   -1   -1    -1    -1
[3,]   -1   -1   -1   -1   -1   -1   -1   -1   -1    -1    -1
[4,]   -1   -1   -1   -1   -1   -1   -1   -1   -1    -1    -1
[5,]   -1   -1   -1   -1   -1   -1   -1   -1   -1    -1    -1
[6,]   -1   -1   -1   -1   -1   -1   -1   -1   -1    -1    -1
[7,]   -1   -1   -1   -1   -1   -1   -1   -1   -1    -1    -1
$corFlag
# Correlation of fishing mortality across ages (0 independent, 1 compound symmetry, or 2 AR(1))
 0
$keyLogFpar
# Coupling of the survey catchability parameters (normally first row is not used, as that is covered
by fishing mortality).
     [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10] [,11]
```

```
[1,]   -1   -1   -1   -1   -1   -1   -1   -1   -1    -1    -1
[2,]    0   -1   -1   -1   -1   -1   -1   -1   -1    -1    -1
[3,]    1   -1   -1   -1   -1   -1   -1   -1   -1    -1    -1
[4,]    2   -1   -1   -1   -1   -1   -1   -1   -1    -1    -1
[5,]    3    4    5    6    7    7   -1   -1   -1    -1    -1
[6,]    8   -1   -1   -1   -1   -1   -1   -1   -1    -1    -1
[7,]    9   -1   -1   -1   -1   -1   -1   -1   -1    -1    -1
$keyQpow
# Density dependent catchability power parameters (if any).
     [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10] [,11]
[1,]   -1   -1   -1   -1   -1   -1   -1   -1   -1    -1    -1
[2,]   -1   -1   -1   -1   -1   -1   -1   -1   -1    -1    -1
[3,]   -1   -1   -1   -1   -1   -1   -1   -1   -1    -1    -1
[4,]   -1   -1   -1   -1   -1   -1   -1   -1   -1    -1    -1
[5,]   -1   -1   -1   -1   -1   -1   -1   -1   -1    -1    -1
[6,]   -1   -1   -1   -1   -1   -1   -1   -1   -1    -1    -1
[7,]   -1   -1   -1   -1   -1   -1   -1   -1   -1    -1    -1
$keyVarF
# Coupling of process variance parameters for log(F)-process (normally only first row is used)
     [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10] [,11]
[1,]    0    0    0    0    0    0    0    0    0     0     0
[2,]   -1   -1   -1   -1   -1   -1   -1   -1   -1    -1    -1
[3,]   -1   -1   -1   -1   -1   -1   -1   -1   -1    -1    -1
[4,]   -1   -1   -1   -1   -1   -1   -1   -1   -1    -1    -1
[5,]   -1   -1   -1   -1   -1   -1   -1   -1   -1    -1    -1
[6,]   -1   -1   -1   -1   -1   -1   -1   -1   -1    -1    -1
[7,]   -1   -1   -1   -1   -1   -1   -1   -1   -1    -1    -1
$keyVarLogN
# Coupling of process variance parameters for log(N)-process
 0 1 1 1 1 1 1 1 1 1 1
$keyVarObs
# Coupling of the variance parameters for the observations.
    [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10] [,11]
[1,]    0    1    2    2    2    2    2    2    2     2     2
[2,]    3   -1   -1   -1   -1   -1   -1   -1   -1    -1    -1
[3,]    4   -1   -1   -1   -1   -1   -1   -1   -1    -1    -1
[4,]    5   -1   -1   -1   -1   -1   -1   -1   -1    -1    -1
[5,]    6    7    8    8    8    8   -1   -1   -1    -1    -1
[6,]    9   -1   -1   -1   -1   -1   -1   -1   -1    -1    -1
[7,]   10   -1   -1   -1   -1   -1   -1   -1   -1    -1    -1
$obsCorStruct
# Covariance structure for each fleet ("ID" independent, "AR" AR(1), or "US" for unstructured). |
Possible values are: "ID" "AR" "US"
 ID ID ID ID AR ID ID
$keyCorObs
# Coupling of correlation parameters can only be specified if the AR(1) structure is chosen above.
# NA's indicate where correlation parameters can be specified (-1 where they cannot).
    1-2 2-3 3-4 4-5 5-6 6-7 7-8 8-9 9-10 10-11
[1,]  NA  NA  NA  NA  NA  NA  NA  NA   NA    NA
[2,]  -1  -1  -1  -1  -1  -1  -1  -1   -1    -1
[3,]  -1  -1  -1  -1  -1  -1  -1  -1   -1    -1
[4,]  -1  -1  -1  -1  -1  -1  -1  -1   -1    -1
[5,]   0   1   1   1   1  -1  -1  -1   -1    -1
[6,]  -1  -1  -1  -1  -1  -1  -1  -1   -1    -1
[7,]  -1  -1  -1  -1  -1  -1  -1  -1   -1    -1
$stockRecruitmentModelCode
# Stock recruitment code (0 for plain random walk, 1 for Ricker, and 2 for Beverton-Holt).
 0
$noScaledYears
# Number of years where catch scaling is applied.
 0
$keyScaledYears
```

```
# A vector of the years where catch scaling is applied.
 Numeric(0)
$keyParScaledYA
# A matrix specifying the couplings of scale parameters (nrow = no scaled years, ncols = no ages).
<0 x 0 matrix>
$fbarRange
# lowest and highest age included in Fbar
 3 7
$keyBiomassTreat
# To be defined only if a biomass survey is used (0 = SSB index; 1 = catch index; 2 = FSB index; 3 =
total catch; 4 = total landings; 5 = TSB index).
 -1  2  2  2 -1 -1 -1
$obsLikelihoodFlag
# Option for observational likelihood | Possible values are: "LN" "ALN"
 LN LN LN LN LN LN LN
$fixVarToWeight
# If weight attribute is supplied for observations this option sets the treatment (0 relative weight,
1 fix variance to weight).
 0
$fracMixF
# The fraction of t(3) distribution used in logF increment distribution
 0
$fracMixN
# The fraction of t(3) distribution used in logN increment distribution
 0 0 0 0 0 0 0 0 0 0 0
$fracMixObs
# A vector with same length as number of fleets, where each element is the fraction of t(3) distribution
used in the distribution of that fleet
0 0 0 0 0 0 0
$constRecBreaks
# Vector of break years between which recruitment is at constant level. The break year is included in
the left interval. (This option is only used in combination with stock recruitment code =3)
numeric(0)
$predVarObsLink
# Coupling of parameters used in a prediction-variance link for observations
     [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10] [,11]
[1,]   -1   -1   -1   -1   -1   -1   -1   -1   -1    -1    -1
[2,]   NA   NA   NA   NA   NA   NA   NA   NA   NA    NA    NA
[3,]   NA   NA   NA   NA   NA   NA   NA   NA   NA    NA    NA
[4,]   NA   NA   NA   NA   NA   NA   NA   NA   NA    NA    NA
[5,]   -1   -1   -1   -1   -1   -1   NA   NA   NA    NA    NA
[6,]   NA   NA   NA   NA   NA   NA   NA   NA   NA    NA    NA
[7,]   NA   NA   NA   NA   NA   NA   NA   NA   NA    NA    NA
$hockeyStickCurve
20
$stockWeightModel
0
$keyStockWeightMean
NA NA NA NA NA NA NA NA NA NA NA
$keyStockWeightObsVar
NA NA NA NA NA NA NA NA NA NA NA
$catchWeightModel
0
$keyCatchWeightMean
NA NA NA NA NA NA NA NA NA NA NA
$keyCatchWeightObsVar
NA NA NA NA NA NA NA NA NA NA NA
$matureModel
0
$keyMatureMean
NA NA NA NA NA NA NA NA NA NA NA
$mortalityModel
```

```
0
$keyMortalityMean
NA NA NA NA NA NA NA NA NA NA NA
$keyMortalityObsVar
NA NA NA NA NA NA NA NA NA NA NA
$keyXtraSd
[,1] [,2] [,3] [,4]
```

Table 17.14. Sole in Division 7.d. SAM model fitting diagnostics of the 2025 assessment.

Model fitting		
log(Lik)	#par	AIC
-487.9265	26	1027.853

Table 17.15. Sole in Division 7.d. SAM model survey catchability of the 2025 assessment.

estimate	1	2	3	4	5	6	7	8	9	10	11
BE-CBT-LPUE	-4.4123	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
UK(E&W)-CBT	-2.5819	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
FR-COTB-model	-4.3931	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
UK(E&W)-BTS-Q3	-7.9948	-7.1636	-7.4916	-7.7378	-7.8837	-7.8837	NA	NA	NA	NA	NA
UK(E&W)-YFS	-9.6638	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
FR-YFS	-11.8197	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

sd	1	2	3	4	5	6	7	8	9	10	11
BE-CBT-LPUE	0.0731	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
UK(E&W)-CBT	0.0682	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
FR-COTB-model	0.0898	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
UK(E&W)-BTS-Q3	0.1996	0.0893	0.0963	0.0785	0.0732	0.0732	NA	NA	NA	NA	NA
UK(E&W)-YFS	0.1352	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
FR-YFS	0.1667	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

Table 17.16. Sole in Division 7.d. Assessment summary.

Year	Recruitment (Age 1)			Spawning stock biomass			Land-ings^	Dis-cards*,^	Fishing pressure (ages 3–7)		
	Low	R	High	Low	SSB	High			Low	F	High
1982	9823	14397	21102	11459	13237	15290	3190	195	0.28	0.34	0.41

Year	Recruitment (Age 1)			Spawning stock biomass			Land-ings^	Dis-cards*,^	Fishing pressure (ages 3–7)		
	Low	R	High	Low	SSB	High			Low	F	High
1983	15212	21872	31448	11970	13712	15708	3458	101	0.30	0.36	0.42
1984	15762	22511	32151	12278	13952	15856	3574	141	0.32	0.37	0.44
1985	11191	16067	23066	12962	14661	16583	3838	242	0.31	0.37	0.43
1986	17283	24761	35472	12606	14216	16031	3932	145	0.35	0.40	0.46
1987	10828	15388	21870	12296	13829	15554	4790	197	0.40	0.46	0.54
1988	19162	26470	36566	11494	12984	14667	3853	198	0.39	0.45	0.53
1989	12739	17694	24575	10141	11378	12766	3805	192	0.41	0.47	0.55
1990	32246	44503	61418	10721	12059	13564	3647	341	0.38	0.43	0.50
1991	25923	35406	48356	11278	12672	14238	4351	368	0.37	0.43	0.50
1992	21478	29729	41149	12744	14516	16534	4072	276	0.34	0.40	0.46
1993	9836	13803	19371	14187	16095	18260	4299	273	0.32	0.38	0.44
1994	20322	27895	38290	13237	14922	16822	4383	123	0.35	0.40	0.47
1995	16872	23242	32016	12639	14101	15732	4420	283	0.38	0.44	0.50
1996	13761	18925	26027	12284	13690	15257	4797	174	0.43	0.50	0.57
1997	19660	27732	39118	11351	12623	14037	4764	148	0.46	0.53	0.61
1998	14241	19632	27063	10463	11662	12997	3363	128	0.43	0.50	0.57
1999	25248	34602	47420	9904	11173	12604	4135	247	0.42	0.48	0.55
2000	28309	38638	52736	10767	12058	13504	3475	202	0.37	0.43	0.50
2001	23571	32599	45085	12786	14431	16289	4025	317	0.35	0.40	0.47
2002	36647	50563	69763	14125	15884	17861	4733	445	0.34	0.39	0.45
2003	17677	24154	33004	16725	18754	21029	6977	583	0.37	0.43	0.50
2004	16545	23160	32419	13360	14913	16647	5819	258	0.37	0.42	0.49
2005	35677	48995	67286	13956	15521	17261	4748	344	0.34	0.40	0.46
2006	30754	41984	57313	13606	15148	16866	4831	315	0.38	0.43	0.50
2007	14745	20566	28684	13681	15337	17194	5420	332	0.40	0.46	0.54
2008	18482	25787	35979	13482	15165	17058	4963	183	0.38	0.45	0.52
2009	31648	44895	63689	12953	14694	16669	4828	287	0.37	0.44	0.52
2010	31258	44044	62062	12684	14384	16312	4108	273	0.34	0.41	0.48

Year	Recruitment (Age 1)			Spawning stock biomass			Land-ings^	Dis-cards*,^	Fishing pressure (ages 3–7)		
	Low	R	High	Low	SSB	High			Low	F	High
2011	25710	36557	51981	13546	15297	17274	4136	342	0.31	0.36	0.43
2012	13493	19201	27322	14214	16093	18221	4057	444	0.28	0.33	0.39
2013	9338	13310	18970	15711	17799	20164	4295	180	0.26	0.31	0.37
2014	10511	14959	21290	14522	16725	19262	4626	216	0.27	0.31	0.37
2015	11656	16909	24531	12707	14672	16940	3385	250	0.25	0.30	0.36
2016	8078	11546	16502	12265	14206	16455	2434	106	0.22	0.27	0.32
2017	13802	19591	27808	10501	12273	14346	2090	156	0.20	0.25	0.30
2018	12425	17644	25053	10215	11984	14061	2395	263	0.21	0.25	0.30
2019	15001	21409	30556	9483	11167	13149	1648	403	0.189	0.23	0.28
2020	6900	10026	14568	9325	11114	13246	1563	409	0.188	0.23	0.28
2021	8438	12215	17682	10067	12095	14531	1567	347	0.182	0.23	0.28
2022	5753	8914	13812	8056	9817	11962	1700	313	0.188	0.23	0.29
2023	8072	14149	24799	8244	10292	12849	1292	207	0.171	0.22	0.27
2024	5806	16044	44336	8204	10370	13107	1112	462	0.160	0.21	0.27
2025		14696**		8127	10542	13704					

*** Discard estimates prior to 2004 assume the average discard proportion by age for 2004–2008 (WKNSEA; ICES, 2021).**

**** Median recruitment resampled (weighted by the inverse of the sd of the recruitment estimates) from the years 2012–2024 (geometric mean shown).**

^ Landings and discards for ages 1–11+ only, as used in the assessment.

Table 17.17. Sole in Division 7.d. Stock numbers for the 2024 SAM assessment (in thousands).

Age	1982	1983	1984	1985	1986	1987	1988	1989	1990	1991	1992	1993	1994
1	14397	21872	22511	16067	24761	15388	26470	17694	44503	35406	29729	13803	27895
2	17883	12476	20039	20842	13815	23480	13107	24850	15043	41754	30338	28665	11616
3	20877	13230	9772	15849	15425	10204	18053	8796	18119	10423	30439	22327	22186
4	5052	13256	7984	5763	9278	8126	5176	9278	4037	10011	5477	17938	13688
5	3217	2990	8309	4729	3557	5430	4200	2864	4702	2072	5519	3188	10759
6	3058	2372	1900	5909	3230	2404	3171	2411	1487	2650	1182	3115	1985
7	1875	2122	1564	1048	3894	2178	1383	1924	1476	978	1572	783	2133
8	921	1207	1417	1011	704	2498	1222	844	1176	945	647	988	552

Age	1982	1983	1984	1985	1986	1987	1988	1989	1990	1991	1992	1993	1994
9	583	621	782	1030	790	502	1595	821	572	802	619	469	678
10	408	414	444	535	799	578	351	1106	556	385	503	400	341
11+	988	1000	1028	1029	1266	1504	1388	1223	1648	1440	1148	1061	1089

Age	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007
1	23242	18925	27732	19632	34602	38638	32599	50563	24154	23160	48995	41984	20566
2	25609	19690	17226	26889	17125	31339	34860	29436	46895	20759	19233	44015	38509
3	9737	19885	14782	13788	23102	12285	22778	24072	19700	29916	14150	12334	30302
4	13408	5675	10154	7040	7224	12536	6025	12894	13109	9683	16029	8389	6816
5	7852	7728	3019	4763	3470	3889	7264	3627	7740	8191	5503	9803	4919
6	6390	4630	4238	1525	2479	1895	2368	4248	2232	4648	4395	3466	5869
7	1453	3985	2799	2463	934	1365	1191	1551	3179	1535	2877	2958	2237
8	1414	1058	2336	1664	1572	546	831	784	1107	1814	1120	1920	1774
9	410	1024	801	1398	1075	943	345	604	591	827	1159	793	1245
10	473	307	677	573	848	661	557	251	464	490	529	791	532
11+	1019	1070	914	1054	1075	1213	1260	1245	1103	1120	1026	1088	1263

Age	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
1	25787	44895	44044	36557	19201	13310	14959	16909	11546	19591	17644	21409	10026
2	17907	23046	38825	40132	33435	16455	11814	13404	15491	9863	18205	15423	19623
3	27576	13588	16074	27118	31421	27577	13500	9405	10286	13329	7659	14087	11729
4	17588	17352	8192	9399	16805	21547	21131	9067	6460	6957	10745	5100	10041
5	3494	9609	9701	4594	5447	10224	14149	13237	5994	4487	4836	7428	3640
6	2919	1968	5325	5631	2943	3526	6958	8914	8052	3969	3228	3521	5151
7	3086	1685	1107	2821	3337	2068	2508	4614	5693	5296	2695	2133	2702
8	1281	1823	910	671	1706	2250	1450	1722	3003	3909	3813	1770	1524
9	1003	810	1090	533	433	1074	1560	982	1175	2084	2856	2694	1286
10	744	684	480	675	317	296	730	1074	742	845	1558	2107	1951
11+	1130	1290	1286	1112	1233	1052	1032	1248	1566	1515	1726	2286	3223

Age	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020

Age	2021	2022	2023	2024
1	12215	8914	14149	16044
2	8485	11001	7665	12603
3	15534	6277	8071	5515
4	7963	11377	4138	5370
5	6912	5335	8182	2789
6	2620	4821	3346	5823
7	3652	1950	3444	2447
8	2100	2863	1326	2644
9	1181	1721	2235	1002
10	994	961	1379	1842
11+	3658	3485	3276	3680

Table 17.18. Sole in Division 7.d. Fishing pressure (F) at age for the 2024 SAM assessment.

Age	1982	1983	1984	1985	1986	1987	1988	1989	1990	1991	1992	1993	1994
1	0.010	0.009	0.009	0.009	0.009	0.009	0.010	0.010	0.011	0.011	0.011	0.010	0.010
2	0.175	0.165	0.173	0.191	0.193	0.208	0.226	0.229	0.235	0.221	0.202	0.185	0.160
3	0.377	0.399	0.436	0.469	0.516	0.552	0.572	0.573	0.523	0.510	0.446	0.429	0.443
4	0.414	0.403	0.413	0.417	0.454	0.500	0.498	0.537	0.504	0.498	0.472	0.449	0.485
5	0.257	0.291	0.289	0.302	0.337	0.402	0.431	0.489	0.462	0.445	0.446	0.430	0.435
6	0.319	0.344	0.366	0.321	0.348	0.430	0.387	0.378	0.337	0.349	0.316	0.289	0.323
7	0.319	0.344	0.366	0.321	0.348	0.430	0.387	0.378	0.337	0.349	0.316	0.289	0.323
8	0.262	0.253	0.233	0.215	0.247	0.271	0.254	0.267	0.282	0.311	0.291	0.271	0.261
9	0.262	0.253	0.233	0.215	0.247	0.271	0.254	0.267	0.282	0.311	0.291	0.271	0.261
10	0.234	0.235	0.235	0.209	0.267	0.292	0.272	0.272	0.290	0.285	0.252	0.226	0.272
11+	0.234	0.235	0.235	0.209	0.267	0.292	0.272	0.272	0.290	0.285	0.252	0.226	0.272

age	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007
1	0.010	0.010	0.010	0.010	0.010	0.010	0.011	0.011	0.011	0.012	0.012	0.013	0.014
2	0.159	0.155	0.153	0.158	0.196	0.217	0.250	0.282	0.293	0.283	0.285	0.251	0.235

age	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007
3	0.514	0.563	0.595	0.577	0.548	0.551	0.500	0.518	0.561	0.506	0.473	0.454	0.415
4	0.505	0.575	0.609	0.577	0.537	0.470	0.421	0.407	0.413	0.440	0.427	0.462	0.509
5	0.476	0.528	0.593	0.532	0.491	0.436	0.440	0.446	0.435	0.472	0.408	0.425	0.445
6	0.349	0.407	0.422	0.401	0.409	0.351	0.324	0.289	0.376	0.353	0.339	0.413	0.477
7	0.349	0.407	0.422	0.401	0.409	0.351	0.324	0.289	0.376	0.353	0.339	0.413	0.477
8	0.265	0.304	0.340	0.349	0.376	0.329	0.237	0.214	0.223	0.293	0.314	0.351	0.395
9	0.265	0.304	0.340	0.349	0.376	0.329	0.237	0.214	0.223	0.293	0.314	0.351	0.395
10	0.294	0.342	0.340	0.351	0.321	0.276	0.249	0.260	0.317	0.388	0.342	0.362	0.373
11+	0.294	0.342	0.340	0.351	0.321	0.276	0.249	0.260	0.317	0.388	0.342	0.362	0.373

age	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
1	0.015	0.015	0.016	0.016	0.017	0.017	0.018	0.019	0.019	0.020	0.021	0.022	0.022
2	0.218	0.210	0.193	0.165	0.141	0.127	0.129	0.134	0.129	0.141	0.158	0.165	0.171
3	0.394	0.389	0.367	0.332	0.300	0.271	0.278	0.267	0.256	0.242	0.260	0.260	0.267
4	0.480	0.481	0.461	0.425	0.392	0.359	0.350	0.313	0.277	0.268	0.269	0.265	0.263
5	0.464	0.424	0.414	0.393	0.356	0.335	0.340	0.319	0.281	0.253	0.248	0.243	0.244
6	0.444	0.456	0.392	0.335	0.305	0.298	0.303	0.301	0.259	0.241	0.235	0.195	0.189
7	0.444	0.456	0.392	0.335	0.305	0.298	0.303	0.301	0.259	0.241	0.235	0.195	0.189
8	0.348	0.378	0.357	0.340	0.308	0.283	0.270	0.235	0.214	0.195	0.189	0.160	0.132
9	0.348	0.378	0.357	0.340	0.308	0.283	0.270	0.235	0.214	0.195	0.189	0.160	0.132
10	0.329	0.331	0.286	0.241	0.224	0.216	0.235	0.243	0.239	0.194	0.175	0.137	0.108
11+	0.329	0.331	0.286	0.241	0.224	0.216	0.235	0.243	0.239	0.194	0.175	0.137	0.108

age	2021	2022	2023	2024
1	0.023	0.024	0.025	0.026
2	0.182	0.200	0.212	0.225
3	0.269	0.284	0.297	0.305
4	0.260	0.256	0.250	0.250
5	0.240	0.251	0.222	0.210
6	0.179	0.187	0.155	0.135

age	2021	2022	2023	2024
7	0.179	0.187	0.155	0.135
8	0.121	0.114	0.091	0.077
9	0.121	0.114	0.091	0.077
10	0.083	0.070	0.054	0.046
11+	0.083	0.070	0.054	0.046

Table 17.19. Sole in Division 7.d. Spawning stock biomass (SSB; tonnes) at age for the 2024 SAM assessment.

age	1982	1983	1984	1985	1986	1987	1988	1989	1990	1991	1992	1993	1994
1	0	0	0	0	0	0	0	0	0	0	0	0	0
2	1583	1117	1859	1977	1296	2140	1035	1989	1395	3474	2412	2203	905
3	4667	3019	2274	3616	3250	2394	4053	1683	4284	2157	6133	4375	4103
4	1707	4327	2713	1759	3073	2590	1560	2726	1267	2893	1588	4805	3482
5	1323	1288	3361	1853	1377	2044	1609	895	1747	782	1702	1138	3298
6	1444	1141	914	2570	1379	1079	1430	953	557	1153	526	1321	778
7	915	1023	723	600	1888	993	704	828	769	458	682	467	900
8	406	548	660	476	317	1044	544	345	474	353	259	473	242
9	284	323	401	592	384	226	855	410	270	337	275	234	318
10	196	251	260	331	434	323	206	628	299	191	242	319	175
11+	711	676	787	886	818	996	989	923	996	874	696	761	721

age	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007
1	0	0	0	0	0	0	0	0	0	0	0	0	0
2	2348	1691	1369	2223	1343	2309	2753	2481	3977	1562	1570	3313	2857
3	1729	3512	2761	2359	4336	2159	4778	4872	4459	4927	2460	2065	4684
4	3347	1406	2564	1791	1678	3033	1851	3540	4103	2092	4032	1941	1440
5	2133	2211	937	1451	959	1377	2227	1221	3018	2105	1927	3005	1346
6	2345	1653	1602	578	769	788	871	1763	1004	1325	1947	1220	1849
7	575	1594	1209	1047	301	643	486	648	582	628	1341	1177	774
8	472	371	818	601	464	202	269	267	455	925	576	893	786
9	195	431	390	656	373	351	164	274	243	323	693	455	617

age	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007
10	243	148	334	259	330	315	296	109	243	252	372	392	211
11+	714	672	639	697	621	882	737	708	671	775	603	687	774

age	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
1	0	0	0	0	0	0	0	0	0	0	0	0	0
2	1338	1808	2654	2531	1843	907	582	725	829	706	1071	850	1009
3	4820	2450	2854	4690	5117	4136	1789	1531	1997	2158	1247	2294	1532
4	4153	4398	2045	2256	3678	5047	4767	2124	1693	1529	2548	1136	1957
5	844	2684	2748	1395	1453	3362	3884	3801	1709	1162	1389	1960	929
6	987	677	1933	1903	974	1199	2407	2745	2665	1210	1046	1074	1406
7	1028	512	402	1047	1291	908	993	1721	2089	1711	924	655	856
8	557	798	338	323	655	936	622	578	1345	1501	1266	623	562
9	374	285	459	218	202	463	690	391	631	996	1060	770	590
10	530	299	260	309	174	123	432	408	338	429	634	760	675
11+	533	782	691	624	706	719	558	648	909	870	800	1044	1597

age	2021	2022	2023	2024
1	0	0	0	0
2	454	531	431	835
3	2472	745	1233	847
4	1682	2119	894	1155
5	1837	1206	1881	695
6	815	1292	1007	1782
7	1209	531	1133	915
8	779	876	518	973
9	470	597	854	503
10	413	338	516	866
11+	1963	1583	1826	1798

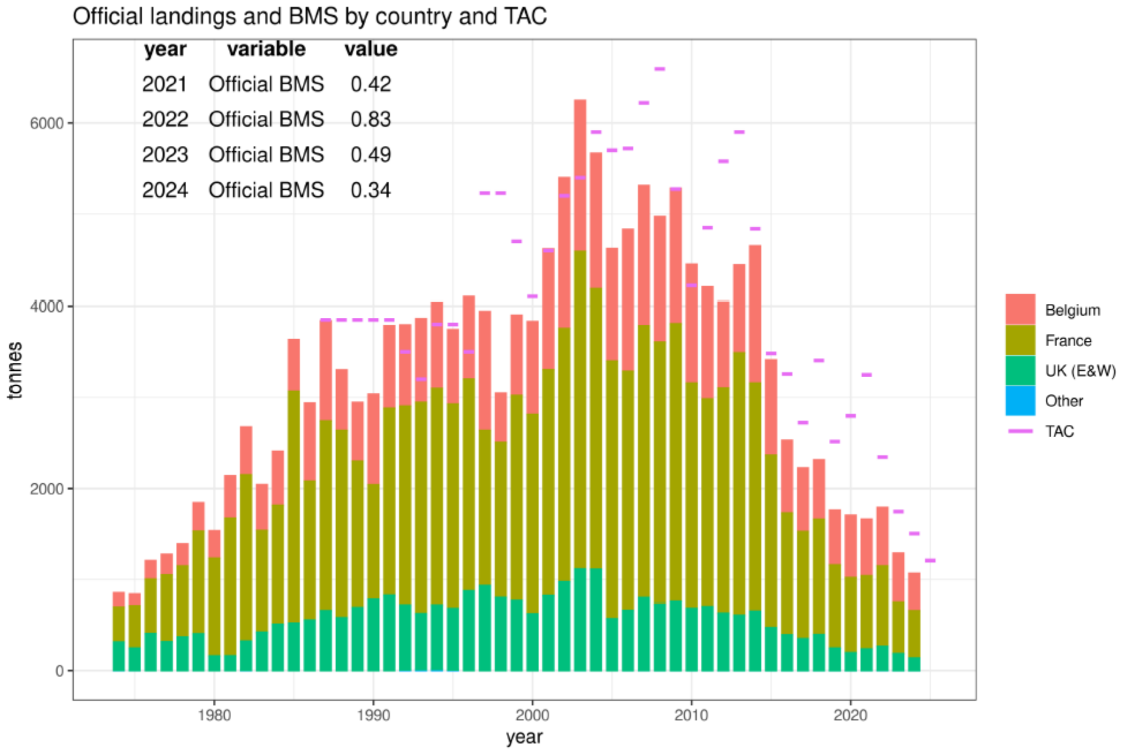


Figure 17.1. Sole in Division 7.d. Official landings (tonnes) by country over the period 1974–2024, as officially reported (Rec 12) (stacked barplot; 'Other' represents landings from e.g. UK Scotland, The Netherlands, Germany or Ireland); pink line represents the TAC (landings; note that from 2016 onwards the TAC represents catch); Official BMS are shown for the last four years in tonnes.

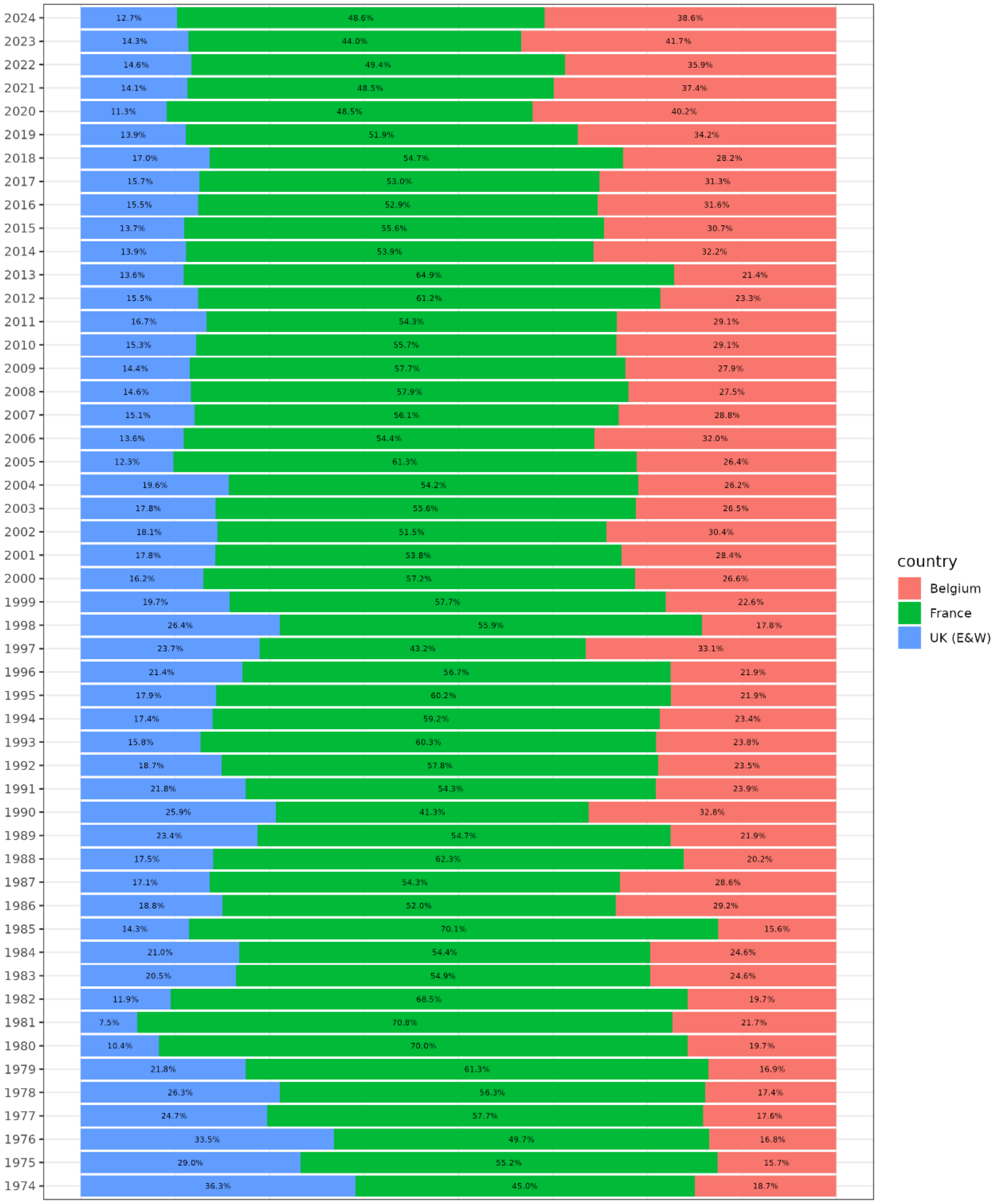


Figure 17.2. Sole in Division 7.d. Relative contribution to the official landings for the main countries involved over the period 1974–2024.

Figure 17.3. Sole in Division 7.d. Uptake of the national quota and total TAC (blue line) in 2022, 2023 and 2024 in comparison with the InterCatch catches, landings and discards (incl. BMS) (bars) and official landings (red line).

Figure 17.4. Sole in Division 7.d. Historic overview (1974–2024) of the official landings (blue line), TAC (pink line) and ICES estimates (InterCatch; including actual discards from 2004 onwards and extrapolated discards prior to 2004; IC catch = red line, IC landings = green line, IC discards (yellow line)); Note that the TAC value represents catch from 2016 onwards.

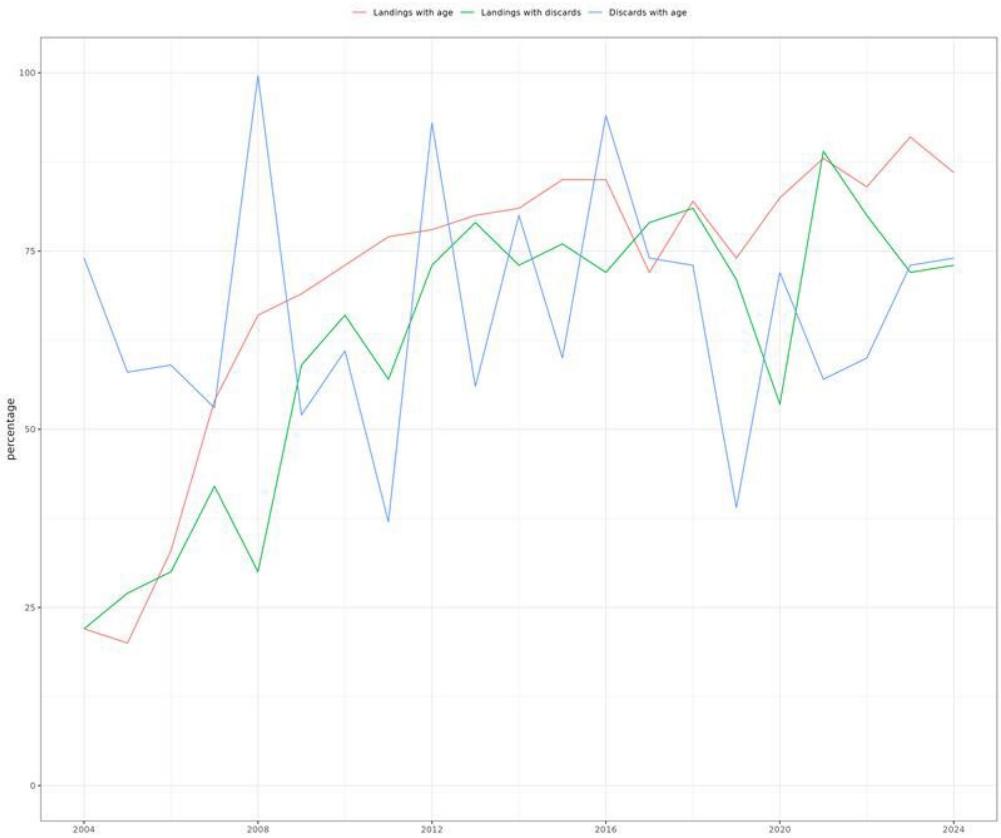


Figure 17.5. Sole in Division 7.d. Overview of data coverage for data uploaded to InterCatch (from 2004 onwards).

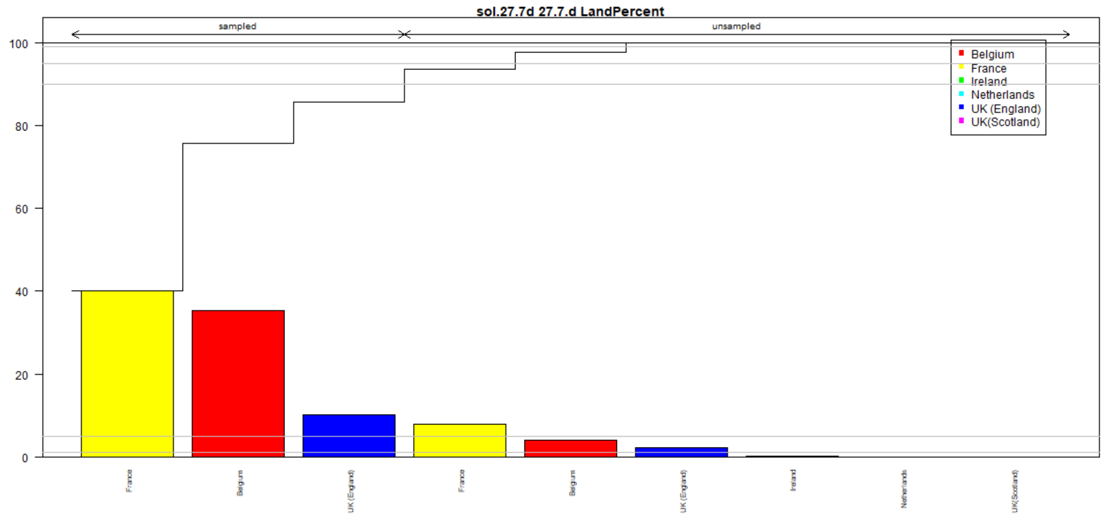


Figure 17.6. Sole in Division 7.d. Overview of the proportion of 2024 landings for which samples (age) have been provided in InterCatch by country.

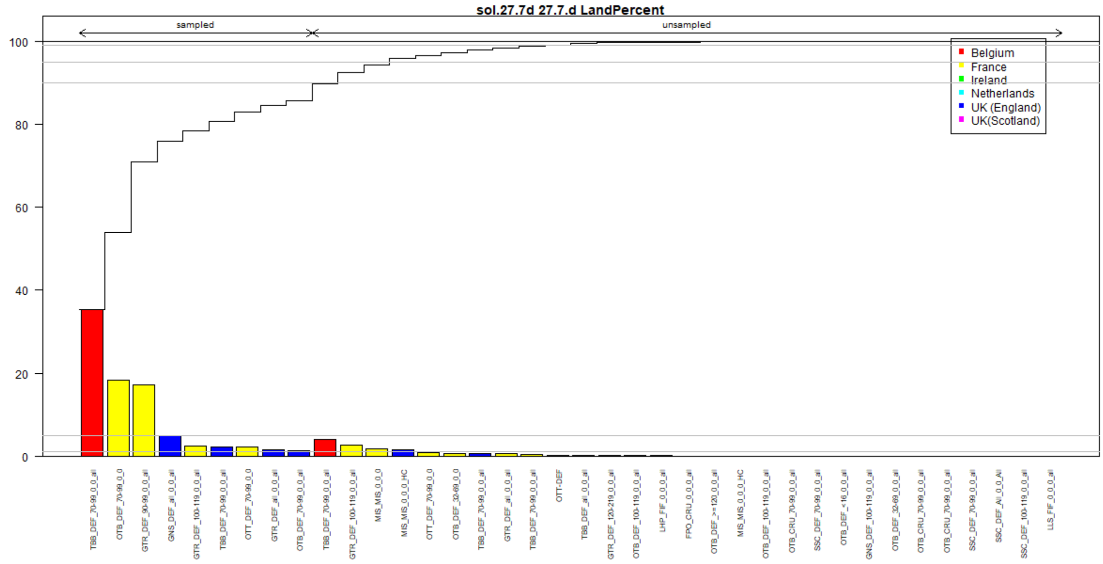


Figure 17.7. Sole in Division 7.d. Overview of the proportion of 2024 landings of sole in Division 27.7.d for which samples have been provided in InterCatch by fleet and country.

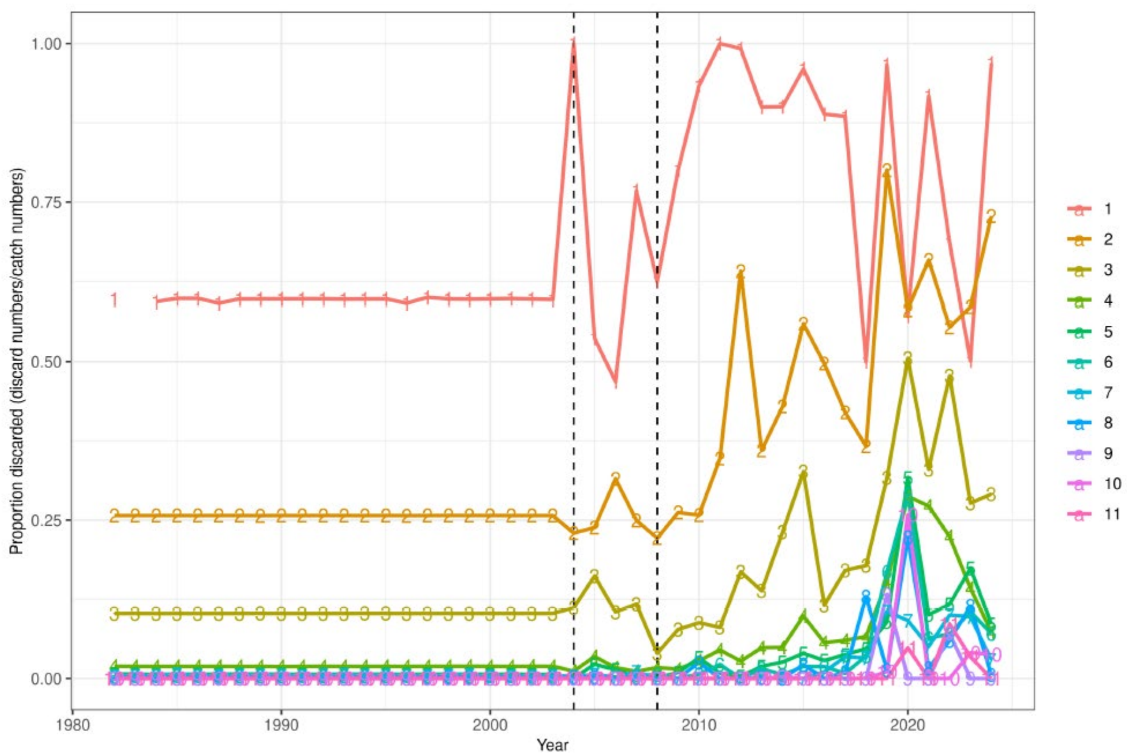


Figure 17.8. Sole in Division 7.d. Proportion discarded (discard numbers at age/catch numbers at age) (data prior to 2004 are estimated using an average discard proportion at age for the period 2004-2008 (indicated by dotted lines)).

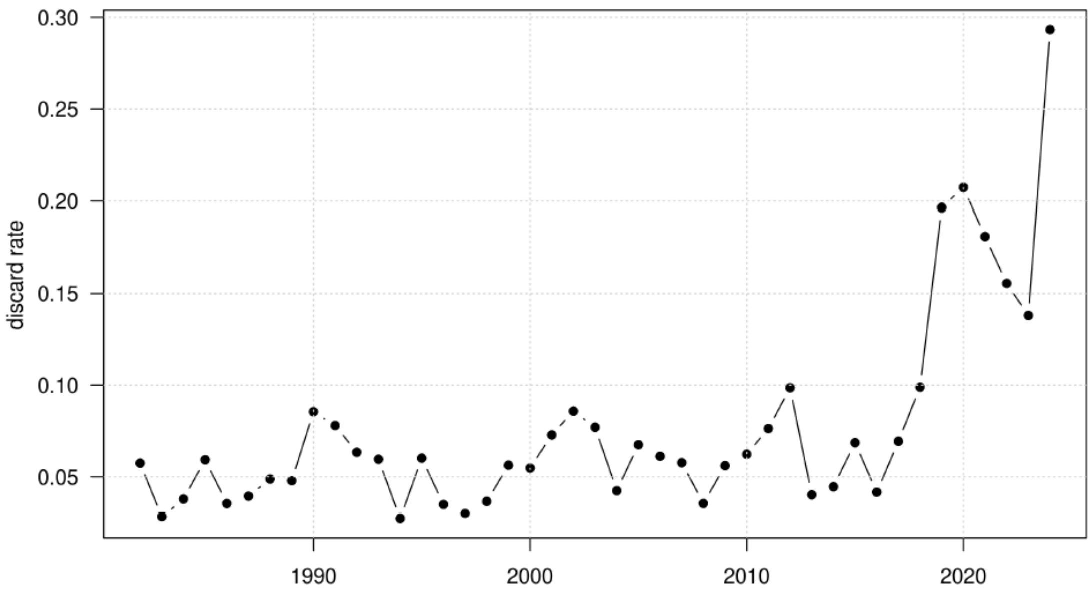


Figure 17.9 Sole in Division 7.d. Discard rate (1982–2024).

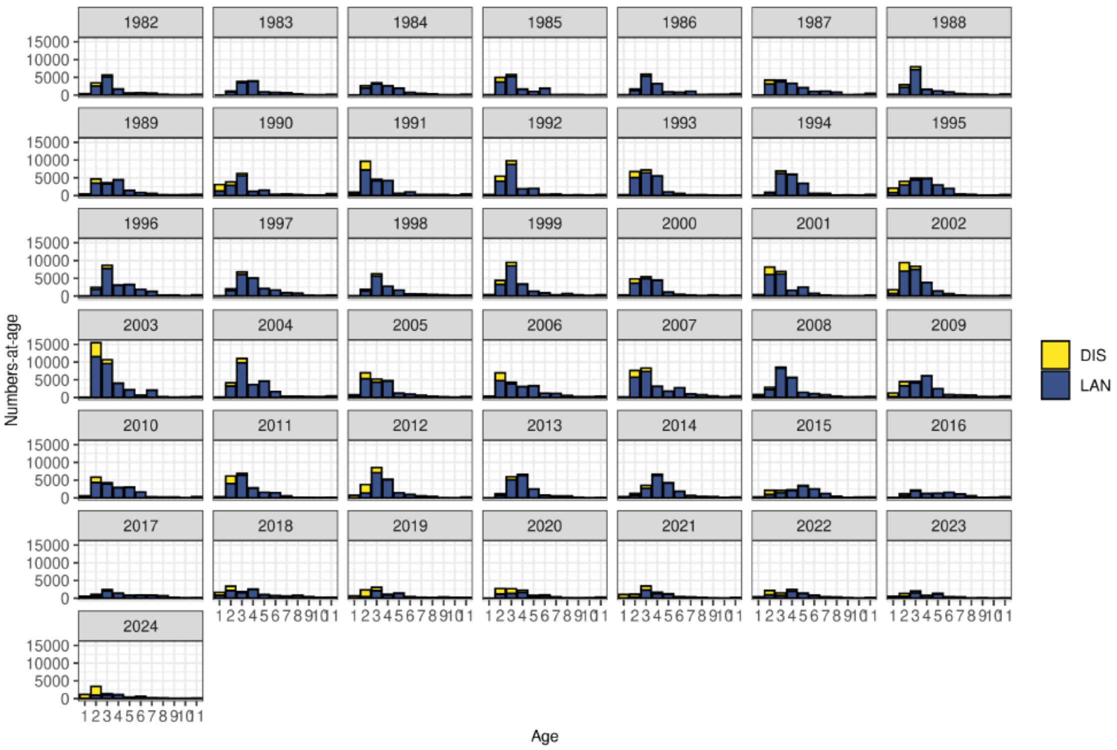


Figure 17.10. Sole in Division 7.d. Landings (blue) and discard (yellow) numbers-at-age over the time-series 1982–2024 for all ages (age 11 is a plusgroup).

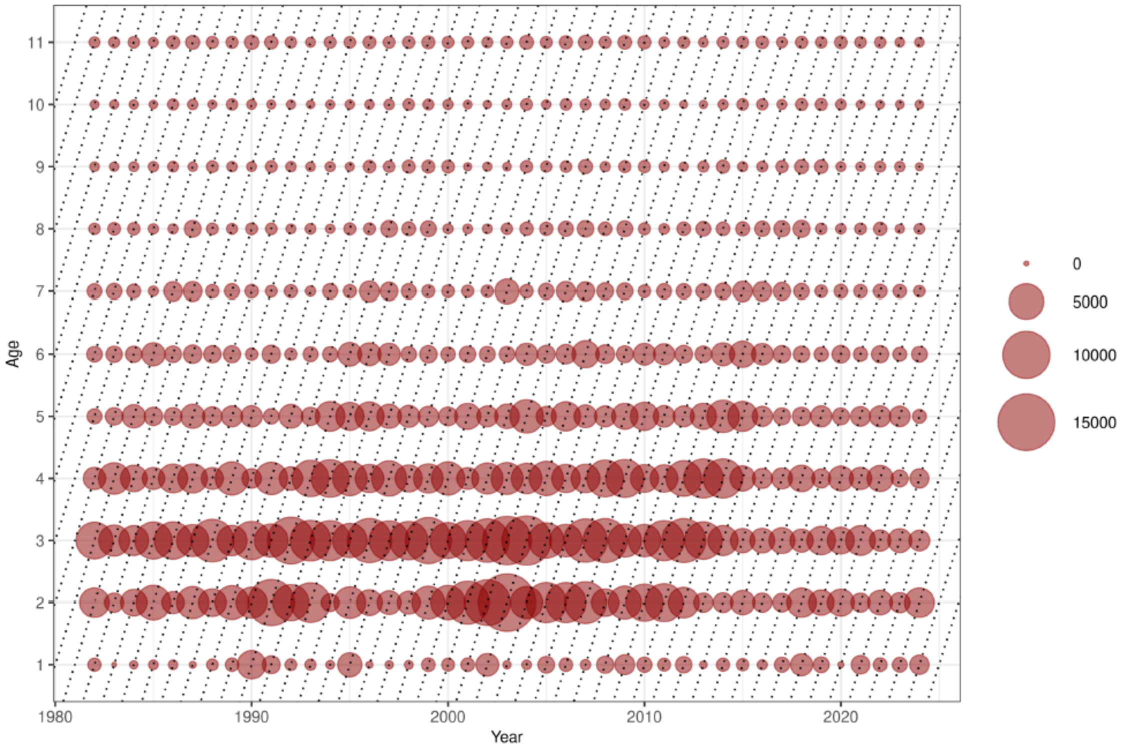


Figure 17.11. Sole in Division 7.d. Catch numbers-at-age over the time-series 1982–2024 for all ages (age 11 is a plusgroup).

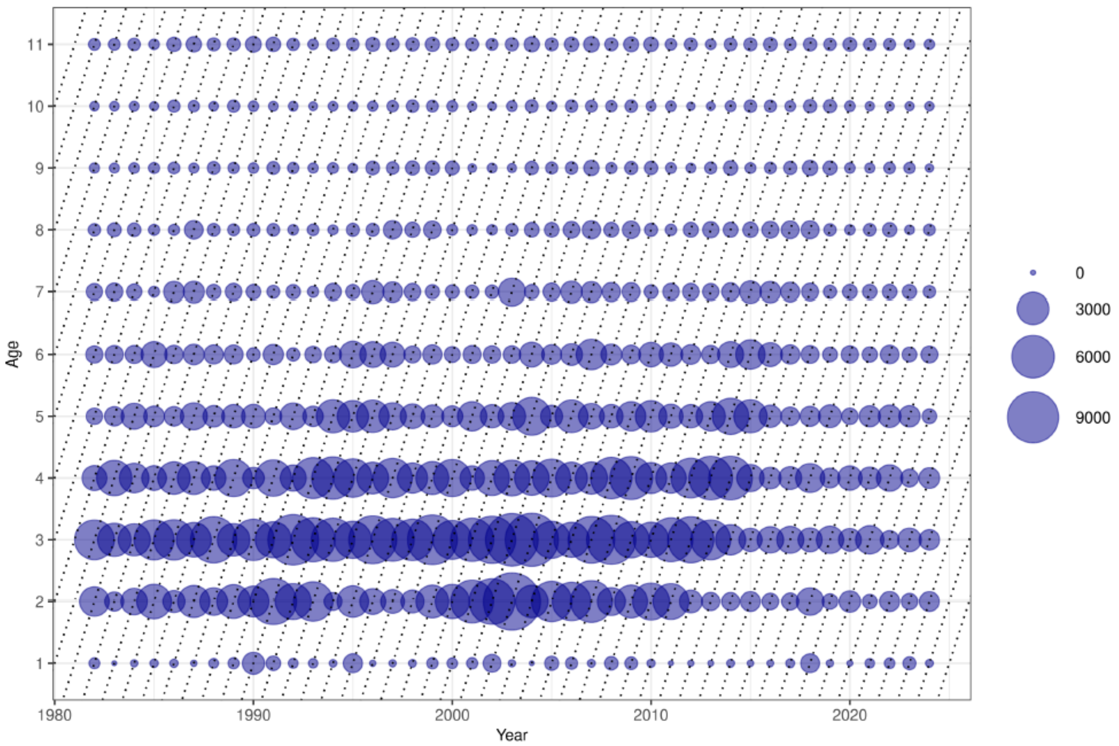


Figure 17.12. Sole in Division 7.d. Landings numbers-at-age over the time-series 1982–2024 for all ages (age 11 is a plusgroup).

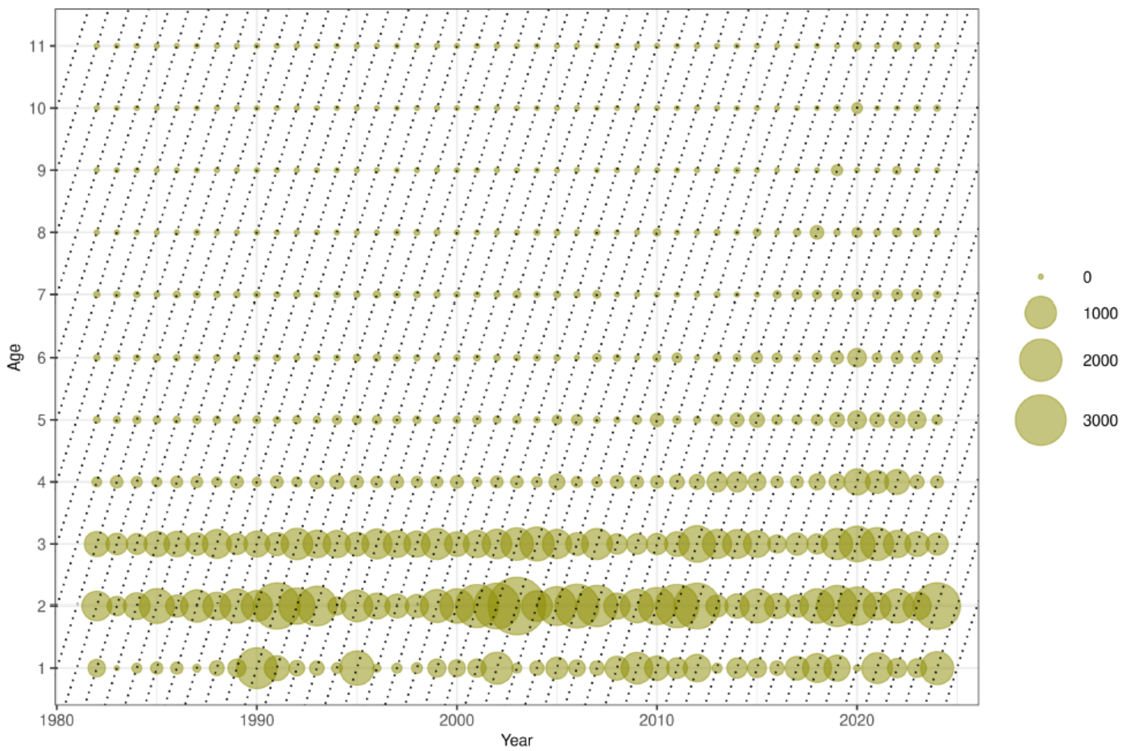


Figure 17.13. Sole in Division 7.d. Discard numbers-at-age over the time-series 1982–2024 for all ages (age 11 is a plusgroup).

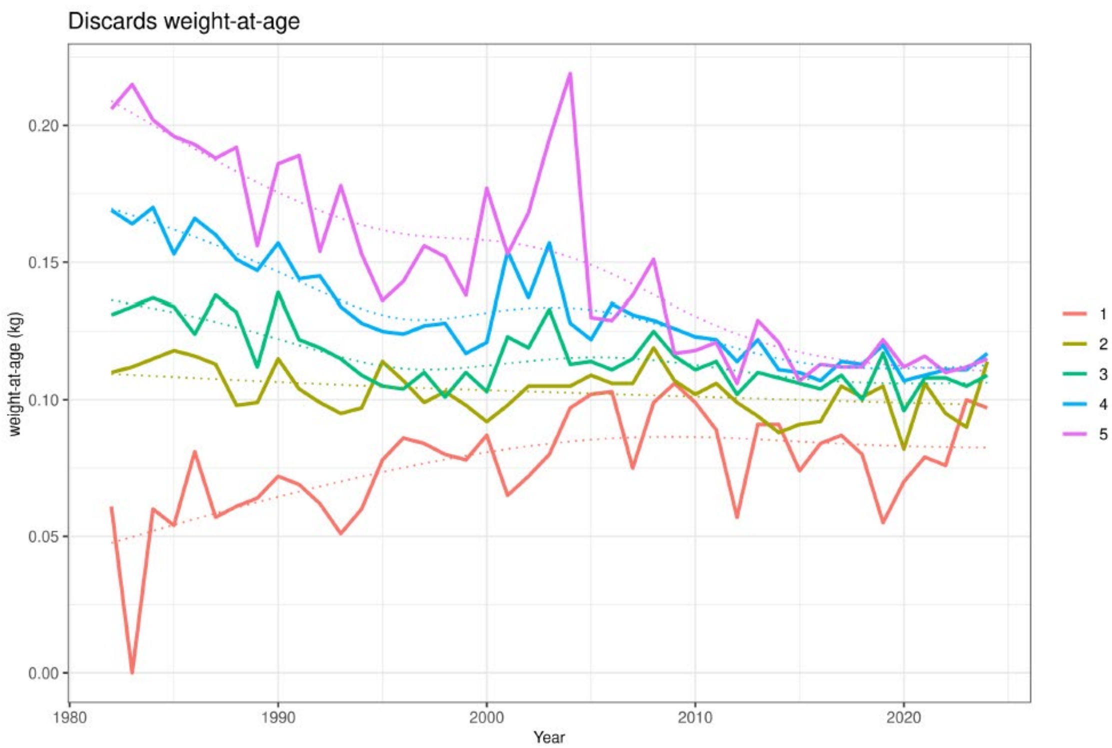


Figure 17.14. Sole in Division 7.d. Discard weights-at-age with smoothed trend line (ages 1–5 are shown).

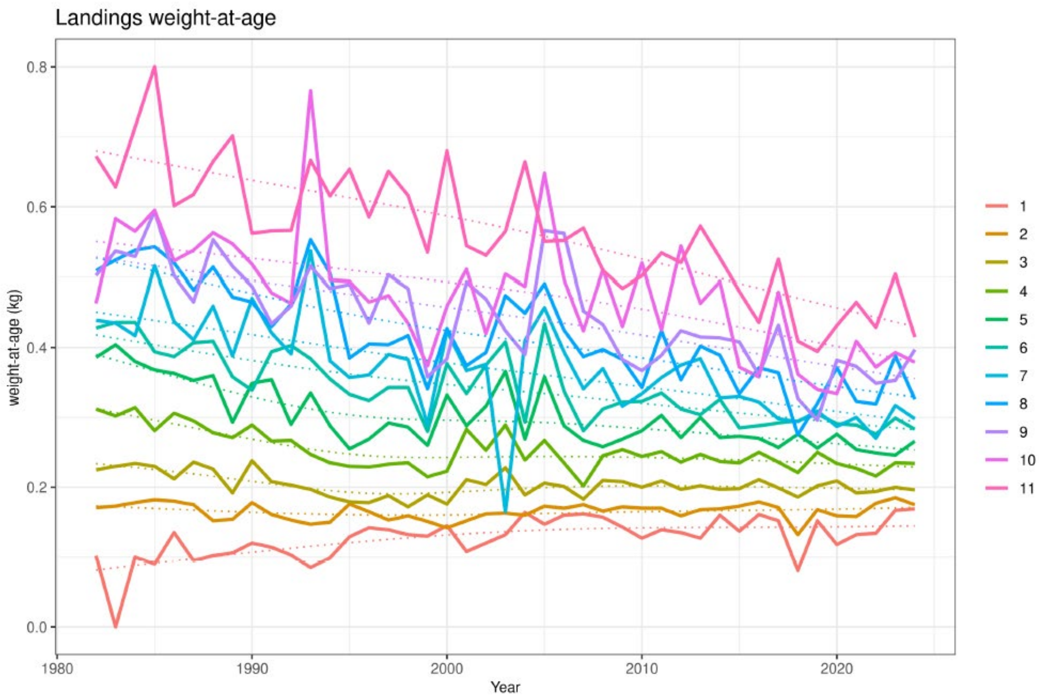


Figure 17.15. Sole in Division 7.d. Landings weights-at-age with smoothed trend line (age 11 is a plusgroup).

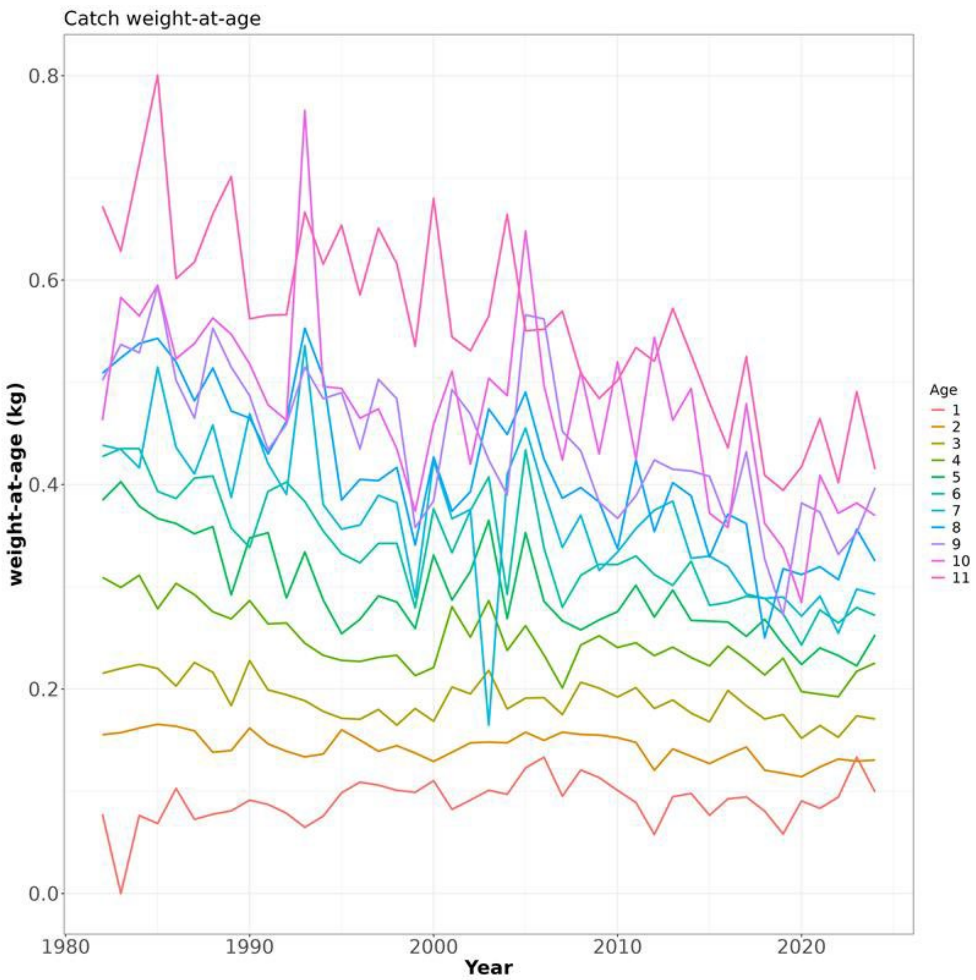


Figure 17.16.

Sole in Division 7.d. Catch weights-at-age (age 11 is a plusgroup).

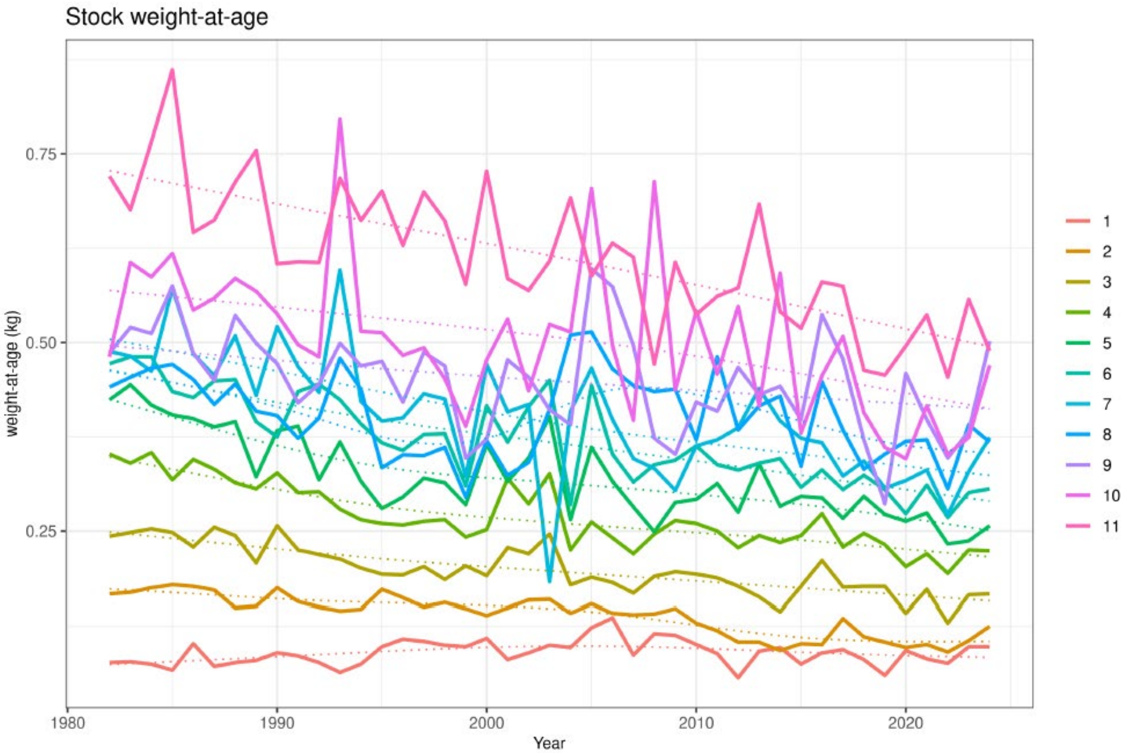


Figure 17.17. Sole in Division 7.d. Quarter 1 stock weights-at-age (kg) (reconstructed for the period 1982–2003 using the ratio of quarter 1 catch weight-at-age and the overall catch weight-at-age for the period 2004–2019 multiplied by the overall catch weight-at-age for 1982–2003).

Figure 17.18. Sole in Division 7.d. Scaled commercial tuning indices: Belgian (red) and UK (blue) commercial beam trawl series and French commercial otter trawl series (green).

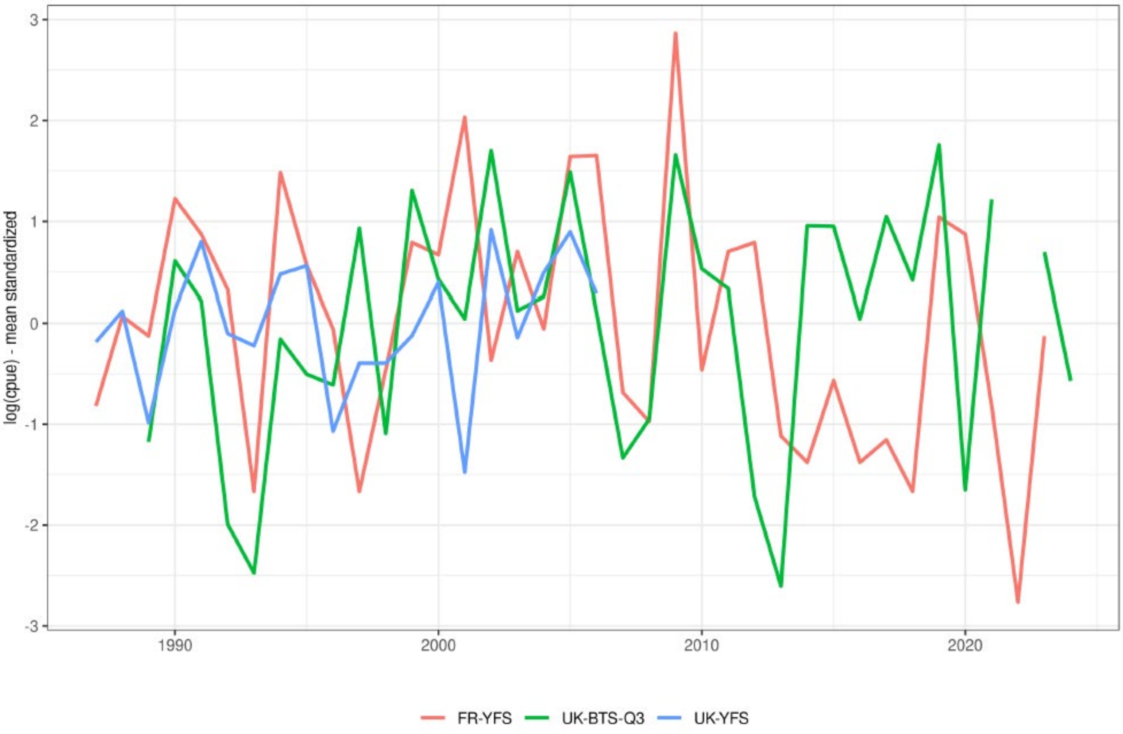


Figure 17.19. Sole in Division 7.d. Scaled survey tuning indices at age: UK (E&W) beam trawl survey (green), UK Young fish survey (blue) and French Young fish survey (red).

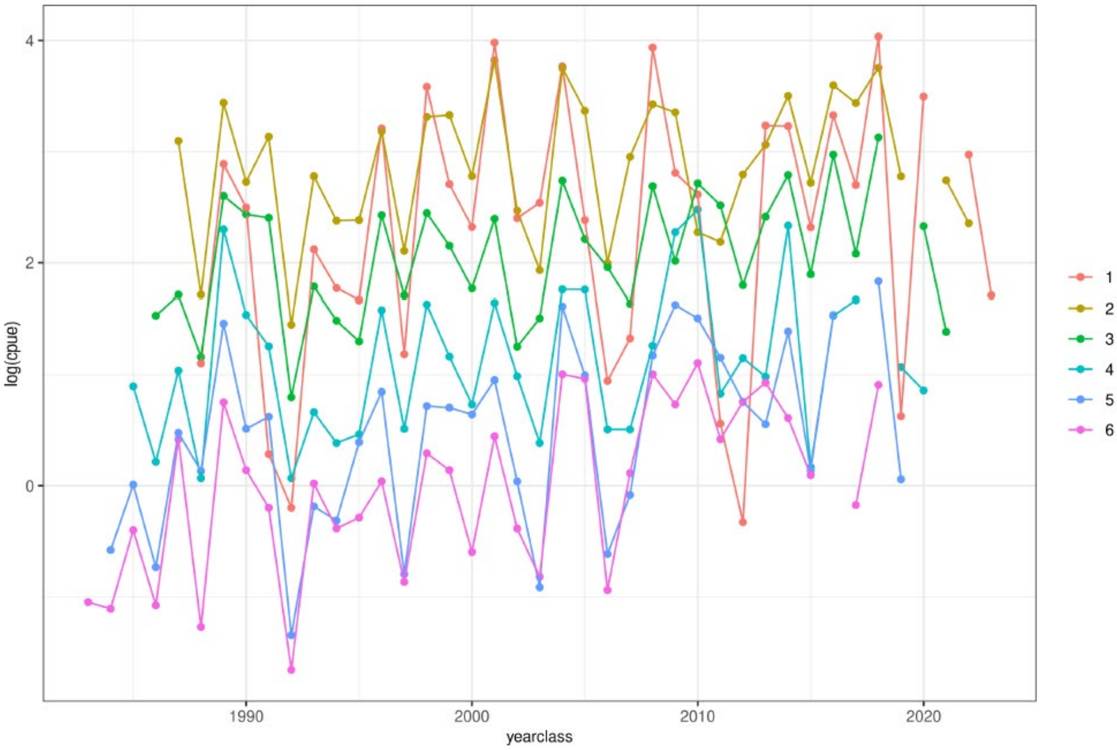


Figure 17.20. Sole in Division 7.d. Log standardised indices of the UK BTS by age and year class.

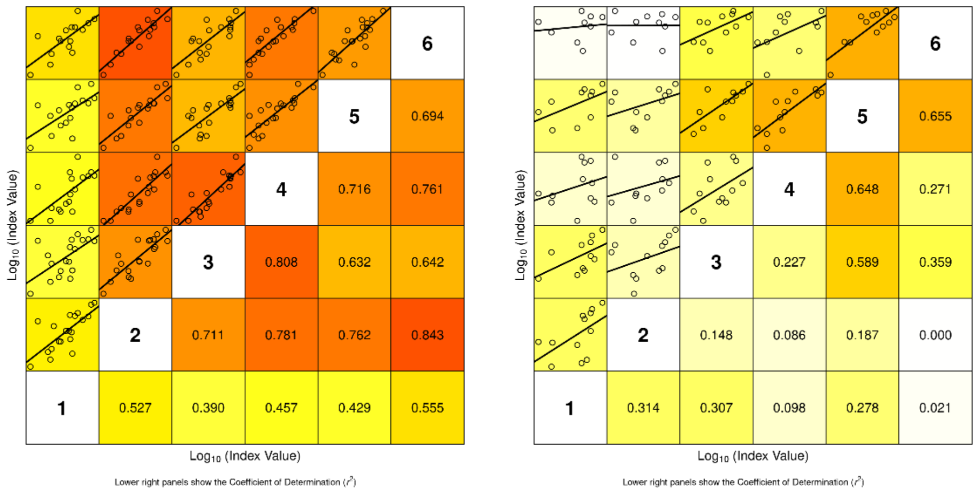


Figure 17.21. Sole in Division 7.d. Internal consistency plot of the UK-BTS tuning series for the early part (1989-2009; left) and last part (2010-2024; right) of the time-series.

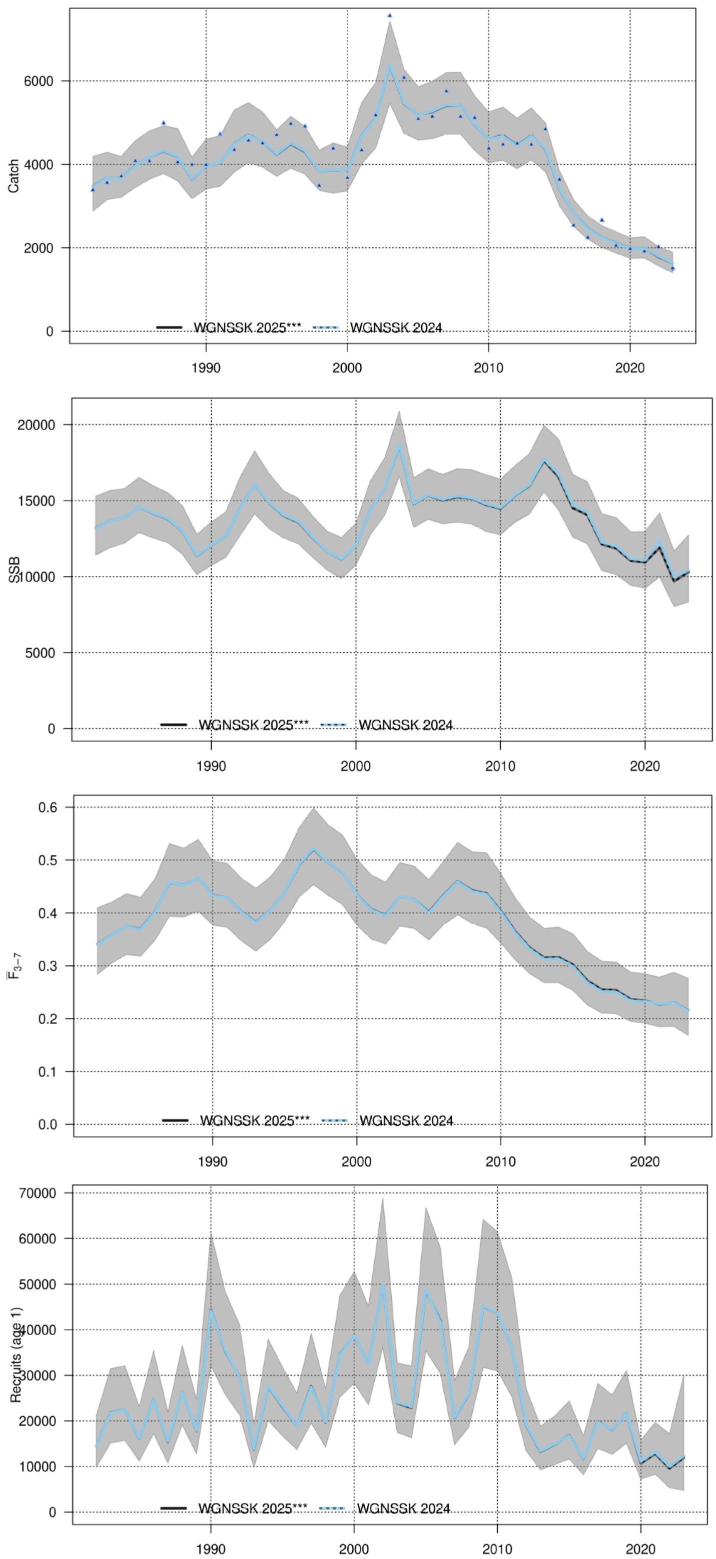


Figure 17.22. Sole in Division 7.d. Summary from last year's assessment (WGNSSK 2024) and an update of last year's assessment (WGNSSK 2025*) that includes updated catch data following a resubmission of UK England for the years 2021 to 2023: trends in catch, spawning stock biomass (SSB), F_{bar} and recruitment (in thousands) are shown with relevant confidence intervals.**

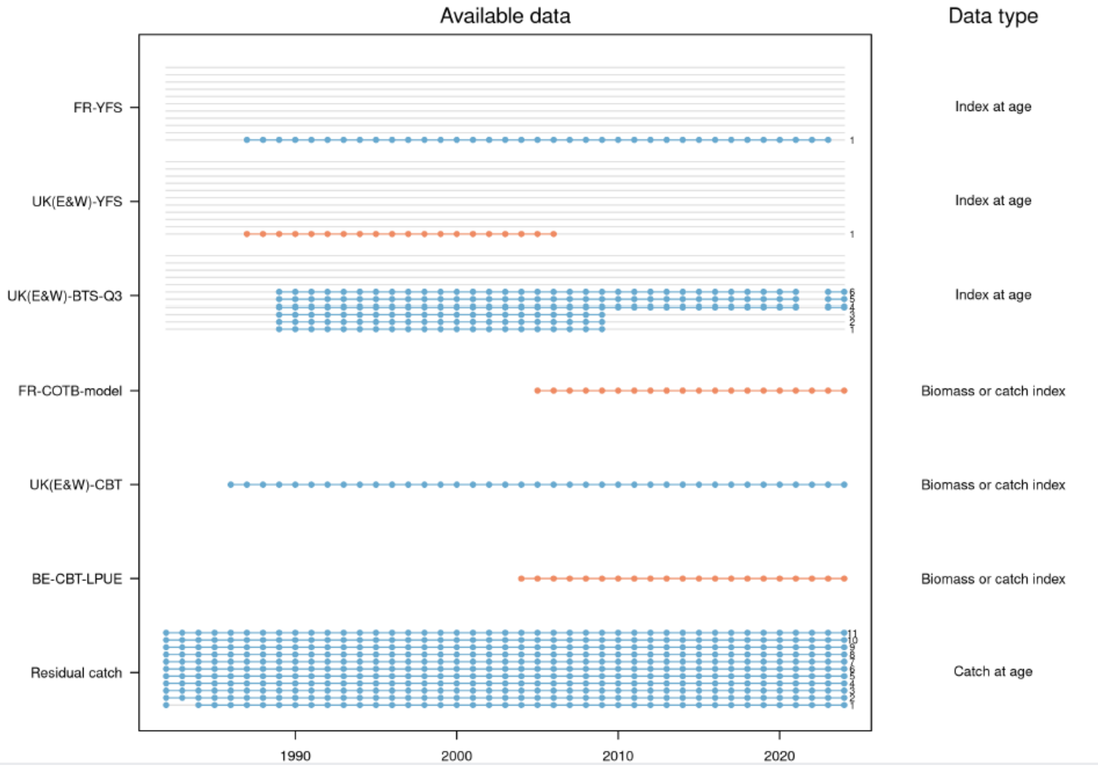


Figure 17.23. Sole in Division 7.d. SAM model input.

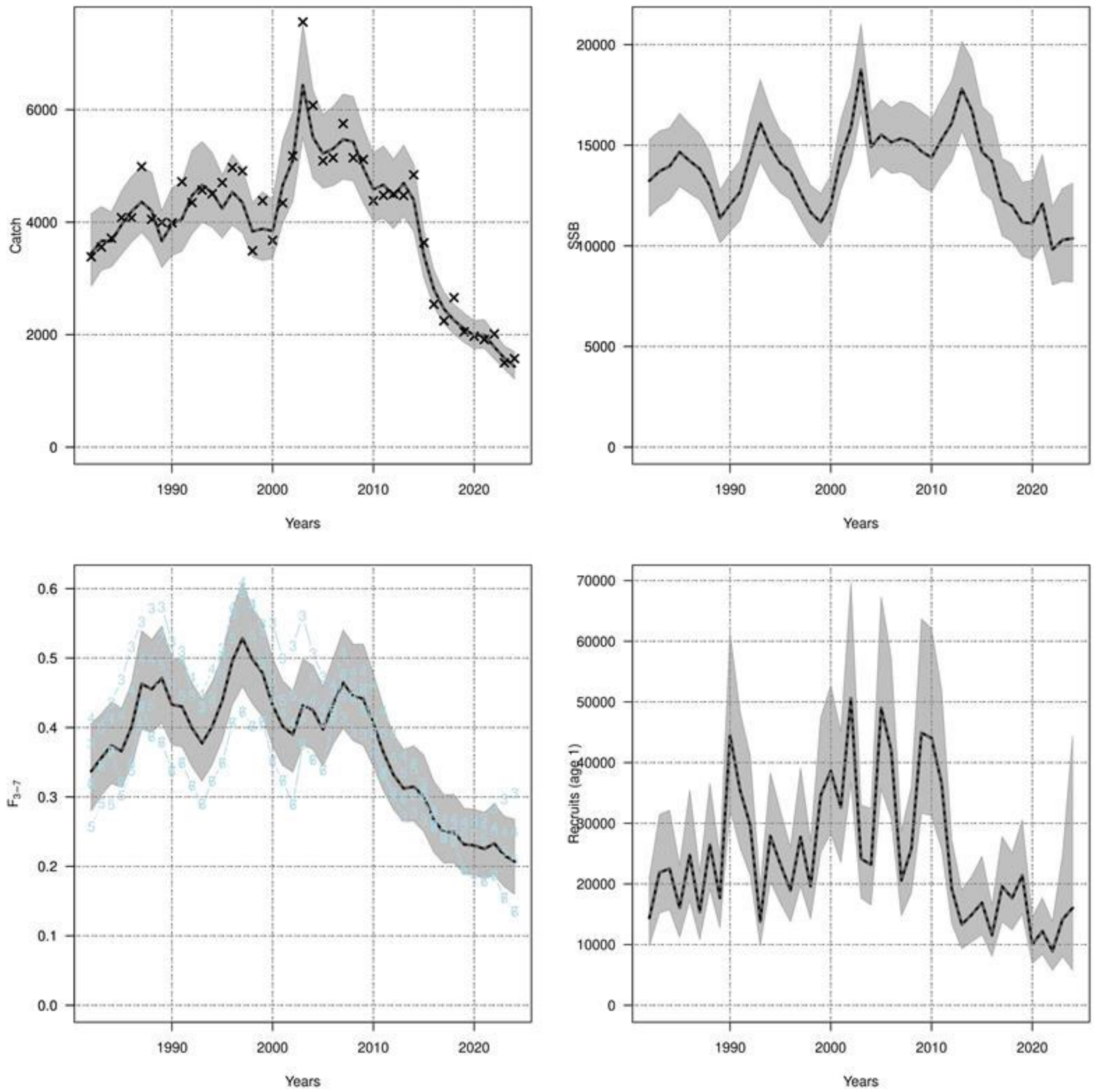


Figure 17.24. Sole in Division 7.d. SAM model summary: trends in catch, spawning stock biomass (SSB), F_{bar} and recruitment (in thousands) are shown with relevant confidence intervals.

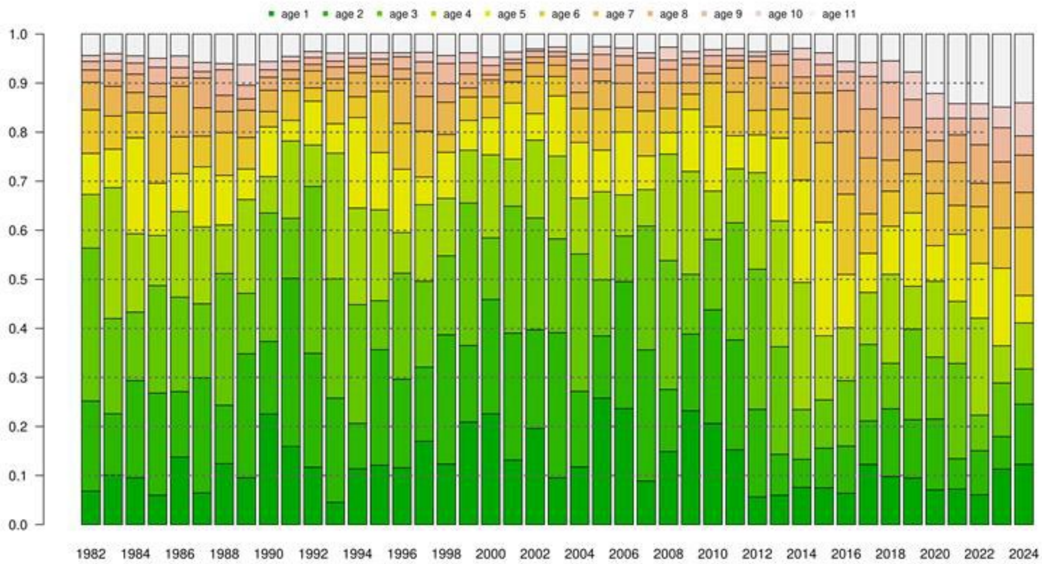


Figure 17.25. Sole in Division 7.d. Proportion of the biomass-at-age.

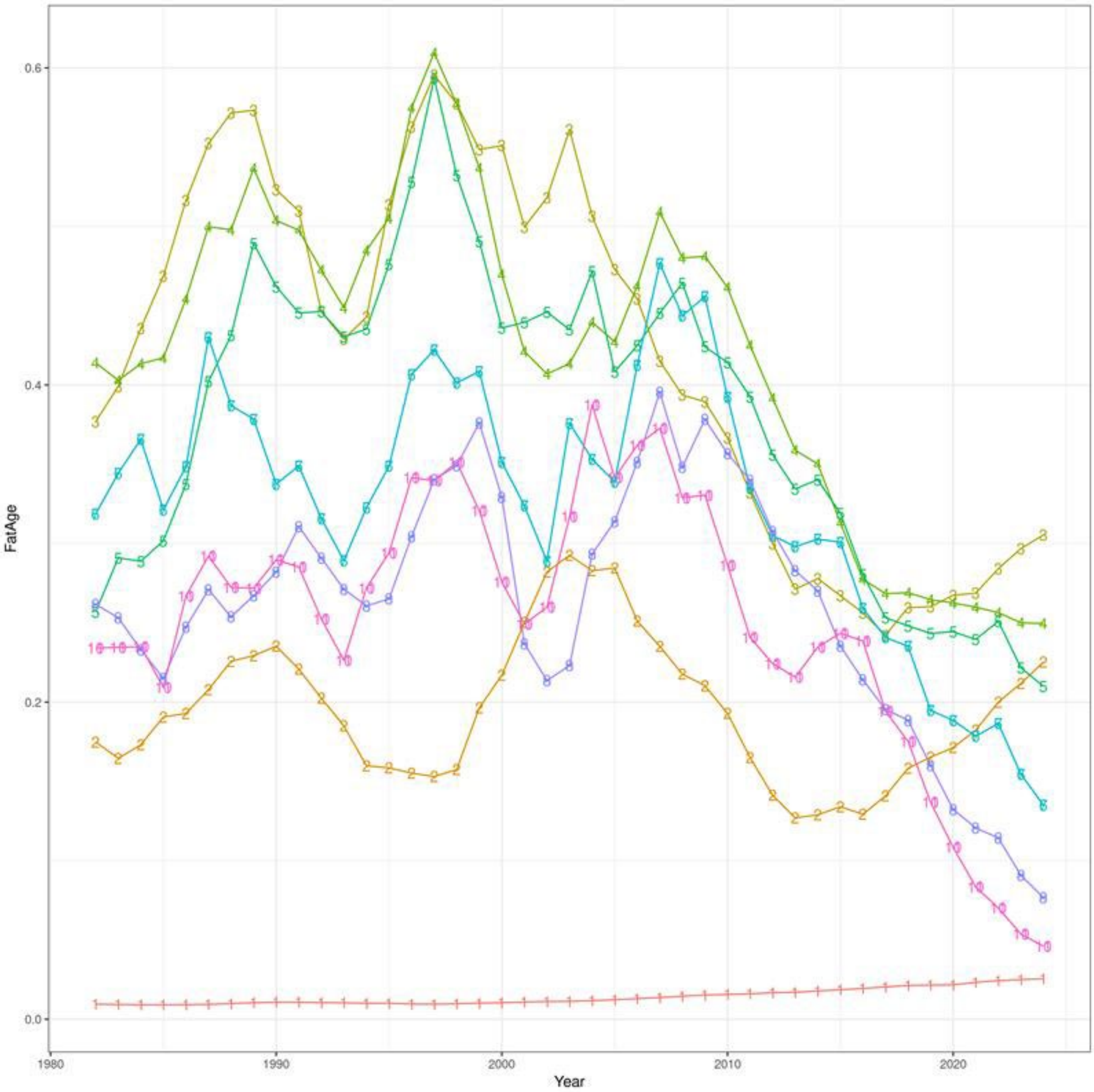


Figure 17.26. Sole in Division 7.d. Fishing pressure at age as estimated by the SAM assessment; Note that age 6 and 7, 8 and 9 and 10 and 11+ overlap.

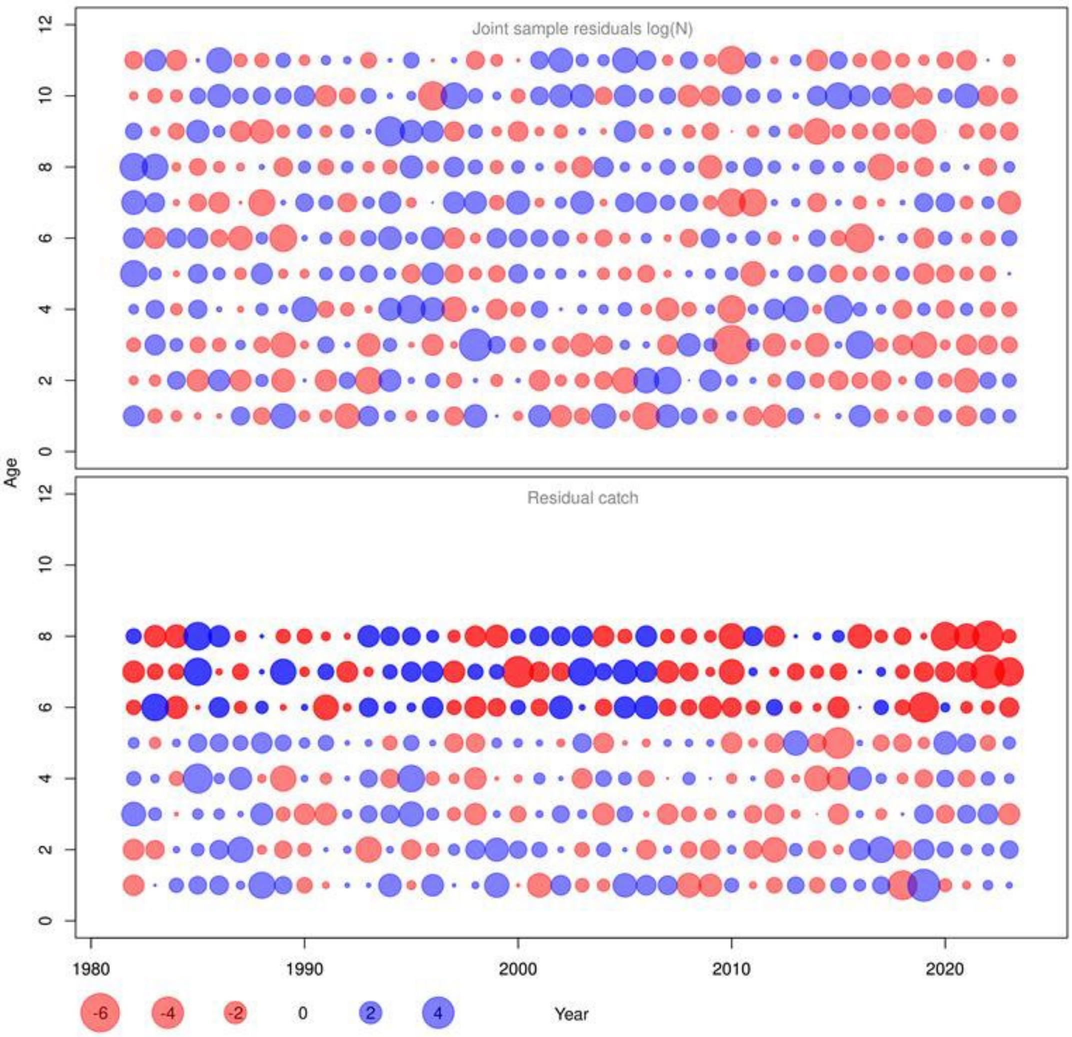


Figure 17.27. Sole in Division 7.d. Process residuals for the survival (logN) and fishing pressure (logF) processes.

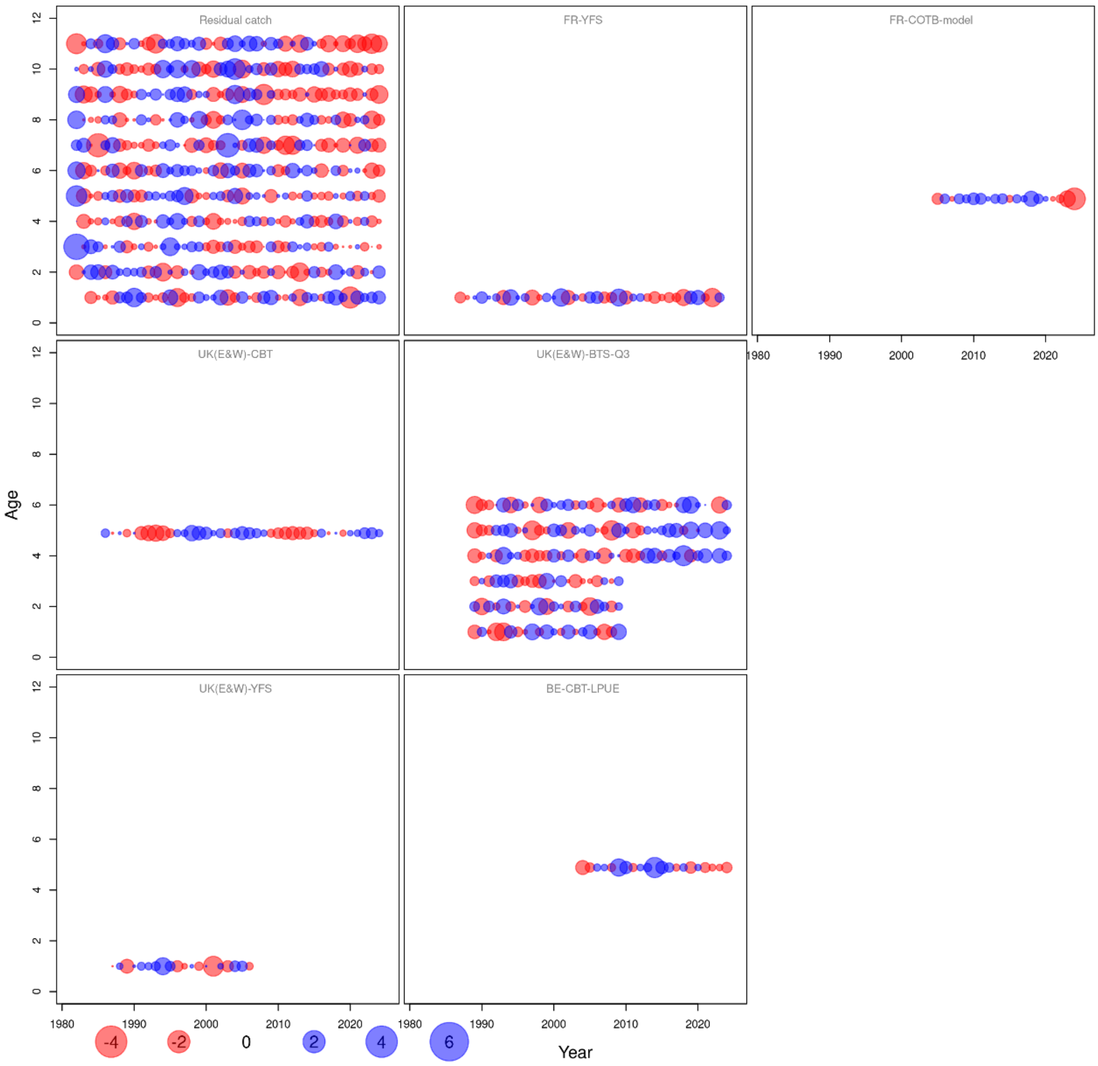


Figure 17.28. Sole in Division 7.d. One step ahead residuals by data stream.

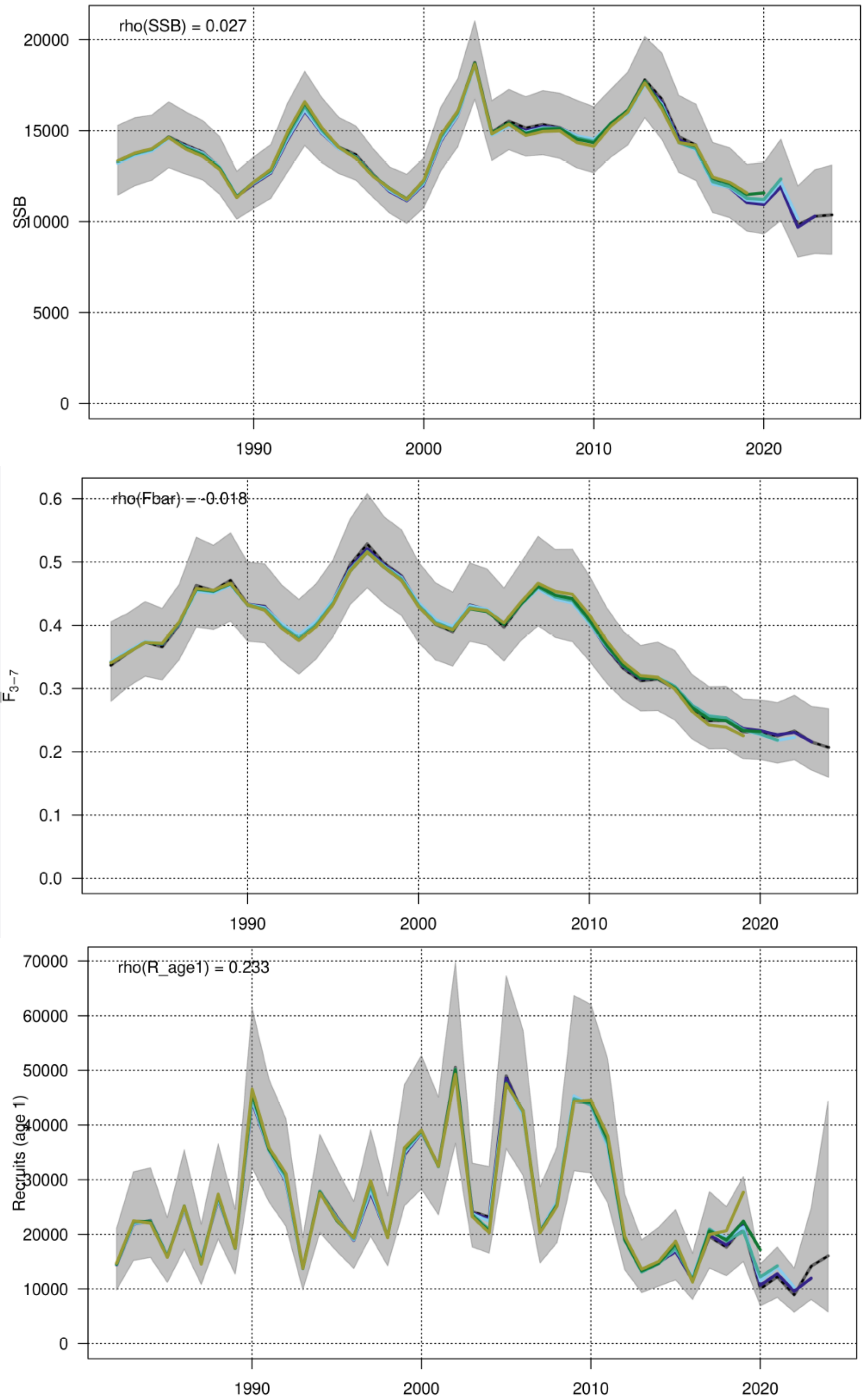


Figure 17.29. Sole in Division 7.d. Retrospective pattern in SSB (Mohn's Rho = 0.027), F_{bar} (Mohn's Rho = 0.018) and recruitment (in thousands; Mohn's Rho = 0.233) based on 5 peels. The grey shades represent the 95% confidence intervals of the model including all data years.

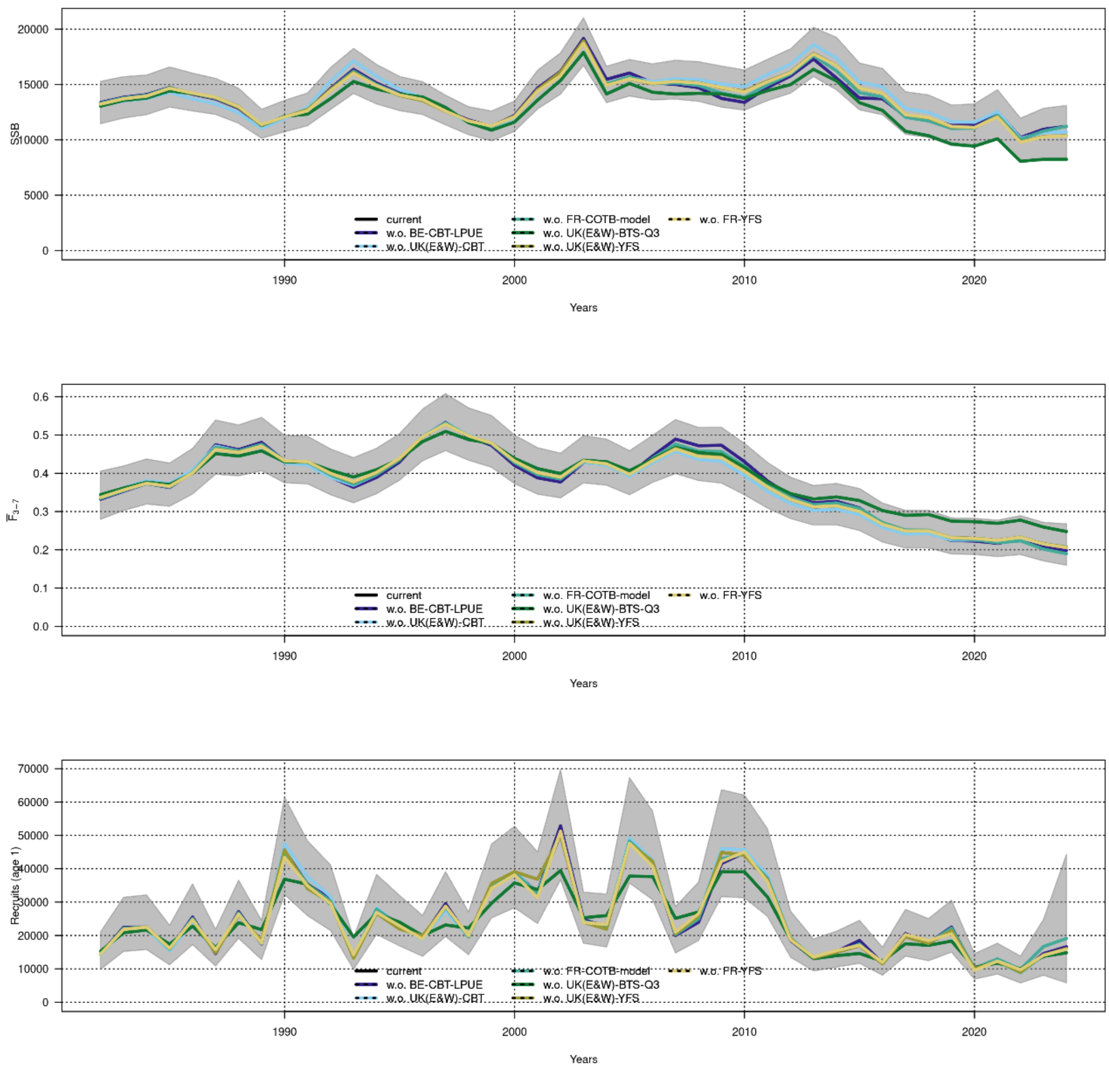


Figure 17.30. Sole in Division 7.d. Leave-one-out analysis. Each coloured line refers to a model fit without the respective tuning fleet. The grey shades represent the 95% confidence intervals of the model including all tuning fleets.

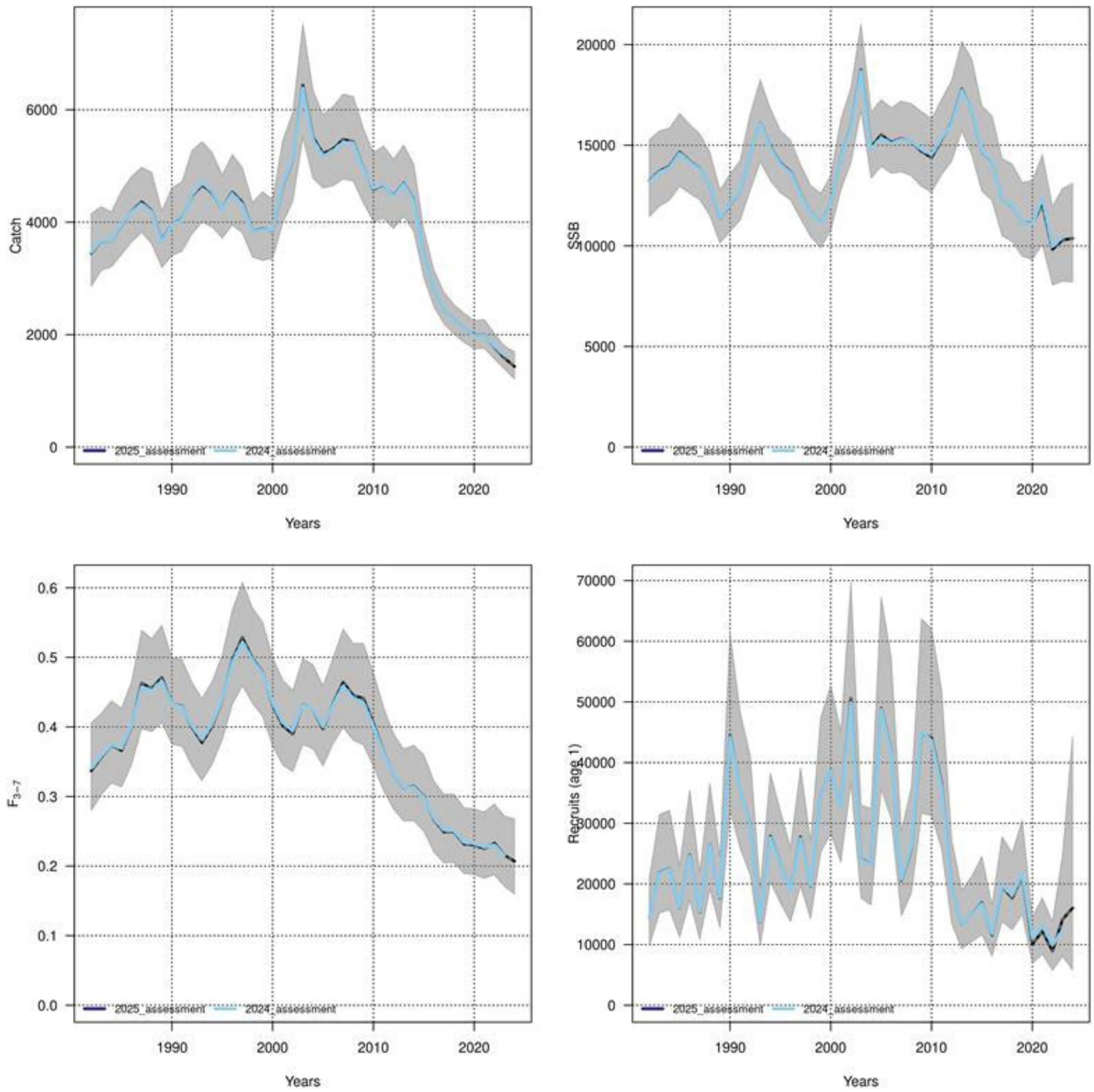


Figure 17.31. Sole in Division 7.d. SAM model summary showing the current model outputs in dark blue (WGNSSK 2025) and the previous assessment model in light blue (WGNSSK 2024). Trends in catch, spawning stock biomass (SSB), F_{bar} and recruitment (in thousands) are shown with relevant confidence intervals.

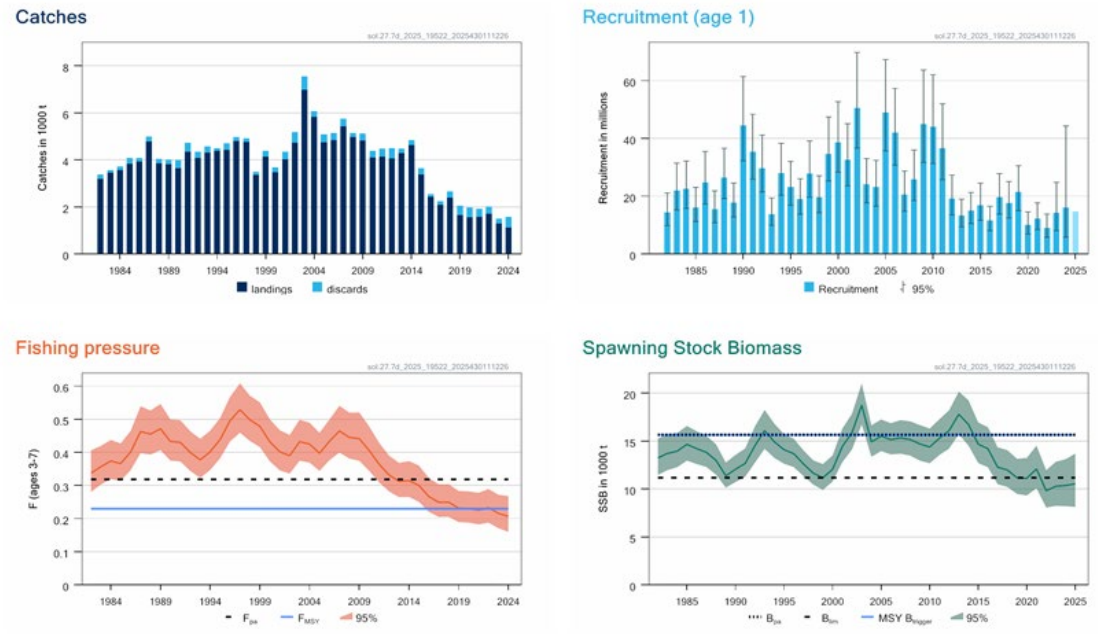


Figure 17.32. Sole in Division 7.d. Summary of the 2025 assessment. The assumed recruitment value for 2025 is shaded in a lighter colour. Discards are reconstructed prior to 2004 and include below minimum size (BMS) landings.

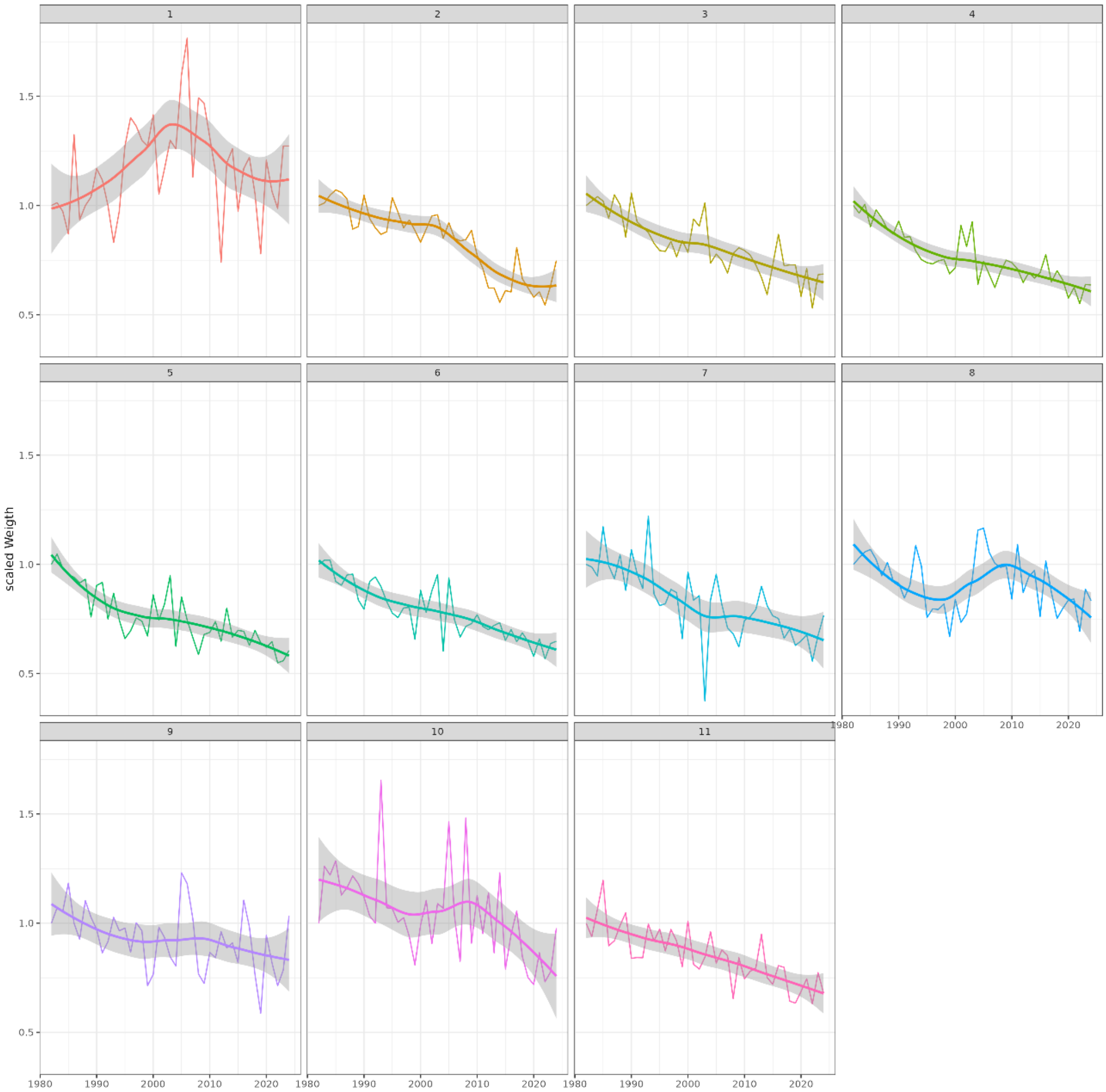


Figure 17.33. Sole in Division 7.d. Development of the stock weights by age group over time scaled to the first value of the time-series. The solid line is a regression spline that aids visualisation of the overall trend.

	1	2	3	4	5	6	7	8	9	10	11
2000	1.001	1.001	0.997	1.003	1.000	1.002	1.004	1.000	0.997	0.994	0.997
2001	1.001	1.000	1.001	0.996	1.004	1.006	1.008	1.010	1.004	1.000	1.002
2002	1.013	1.001	0.998	1.004	1.000	1.007	1.012	1.013	1.016	1.012	1.008
2003	1.005	1.011	0.999	1.001	1.007	1.002	1.009	1.017	1.018	1.019	1.011
2004	1.002	1.008	1.010	0.999	1.004	1.007	1.000	1.007	1.020	1.018	1.012
2005	1.005	1.007	1.010	1.016	1.004	1.005	1.009	1.002	1.009	1.021	1.012
2006	1.005	1.001	0.999	1.004	1.007	1.004	1.004	1.008	1.002	1.008	1.017
2007	0.992	1.003	1.003	0.998	1.001	1.006	1.000	1.000	1.003	0.999	1.010
2008	0.990	0.994	1.000	1.002	0.992	0.999	0.997	0.992	0.994	0.997	1.003
2009	0.993	0.990	0.993	0.995	0.996	0.989	0.995	0.993	0.991	0.992	0.997
2010	1.009	0.991	0.990	0.990	0.988	0.992	0.986	0.988	0.989	0.986	0.991
2011	1.014	1.009	0.997	0.994	0.990	0.992	0.991	0.986	0.986	0.986	0.989
2012	1.002	1.013	1.012	1.002	0.997	0.993	0.994	0.992	0.985	0.985	0.991
2013	0.997	1.004	1.012	1.009	1.001	0.998	0.994	0.994	0.990	0.985	0.991
2014	1.002	0.998	1.003	1.003	1.000	0.997	0.996	0.991	0.988	0.986	0.986
2015	1.011	1.001	1.000	1.003	1.003	0.999	0.995	0.993	0.986	0.982	0.980
2016	0.982	1.010	1.002	1.001	1.006	1.003	0.998	0.993	0.991	0.981	0.975
2017	0.991	0.983	1.011	1.004	1.004	1.012	1.007	1.000	0.992	0.988	0.973
2018	0.975	0.988	0.979	1.008	1.004	1.003	1.017	1.006	0.997	0.988	0.974
2019	0.989	0.974	0.993	0.980	1.022	1.013	1.005	1.029	1.010	0.998	0.982
2020	0.904	0.990	0.971	1.002	0.978	1.046	1.023	1.006	1.045	1.014	0.997
2021	0.931	0.898	0.991	0.967	1.019	0.970	1.083	1.032	0.998	1.063	1.019
2022	0.895	0.930	0.880	0.986	0.964	1.045	0.971	1.148	1.039	1.009	1.024
2023	1.147	0.879	0.915	0.862	0.984	0.953	1.065	0.946	1.164	1.045	1.005
2024	1.074	1.130	0.837	0.883	0.812	0.971	0.919	1.076	0.892	1.186	0.937
2025	1.002	1.054	1.110	0.820	0.876	0.808	0.996	0.934	1.085	0.904	1.008

Figure 17.34. Sole in Division 7.d. Ratio of the numbers at age of this year's assessment and forecast against last year's assessment and forecast. Blue cells indicate an overestimation of last year's stock numbers, while red cells indicate an underestimation.

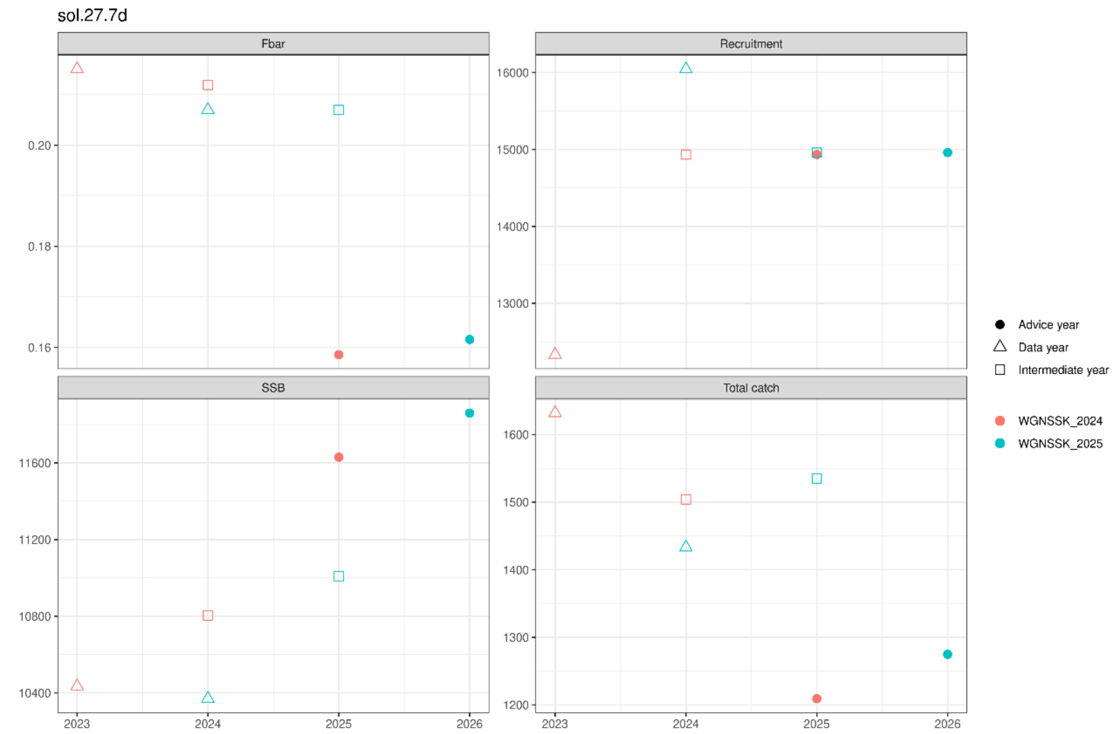


Figure 17.35. Sole in Division 7.d. Comparison of last year's estimates and forecast assumption of SSB, F_{bar}, recruitment and catch against this year's estimates and forecast assumptions.

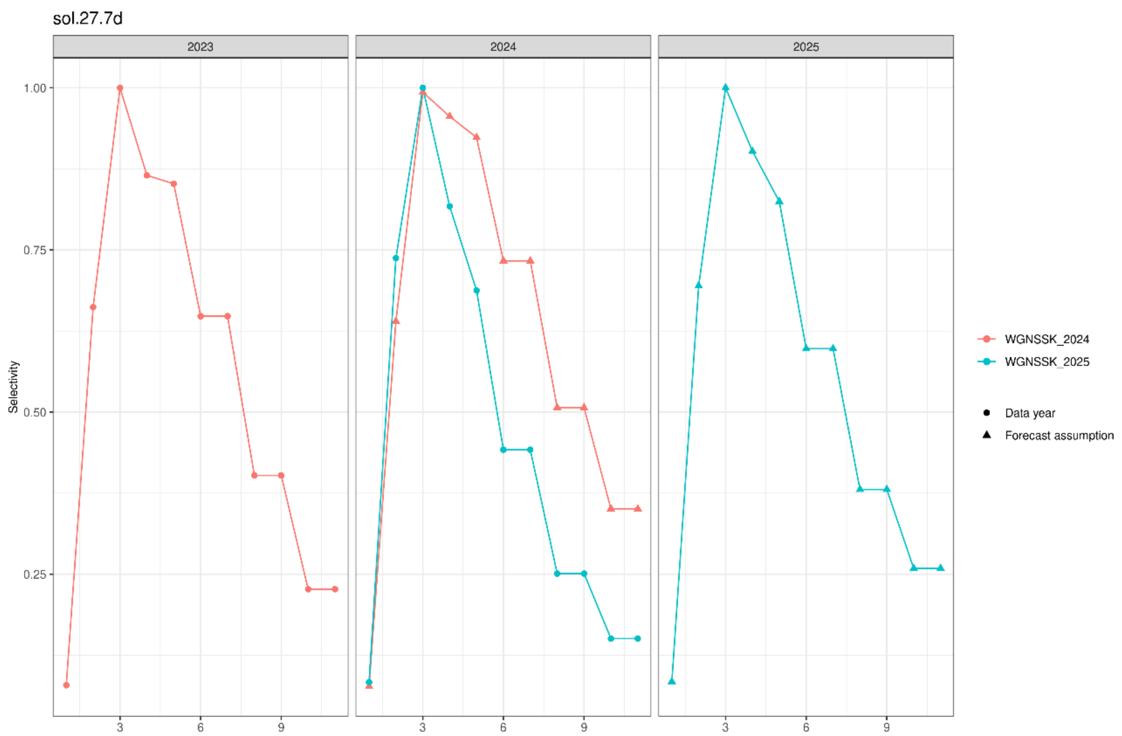


Figure 17.36. Sole in Division 7.d. Comparison of last year's selectivity estimates and assumption against this year's assessment.

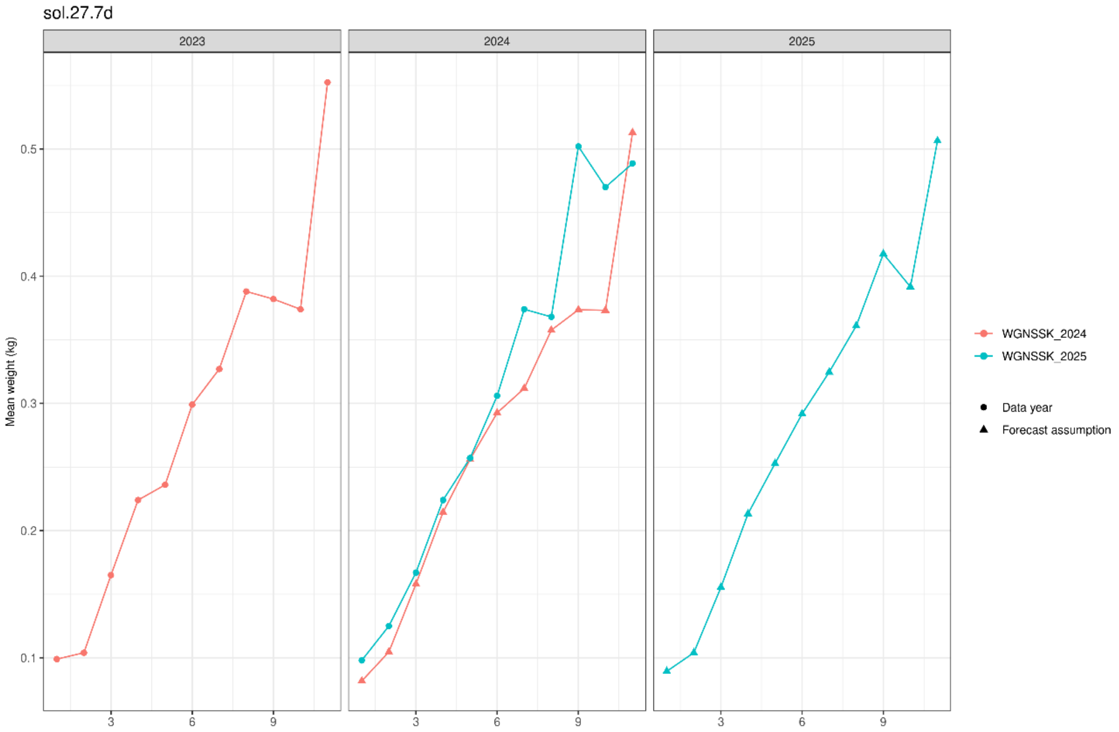


Figure 17.37. Sole in Division 7.d. Comparison of last year's stock weights at age (observed and forecast assumption) against the stock weights at age (observed and forecast assumption) of this year's assessment.

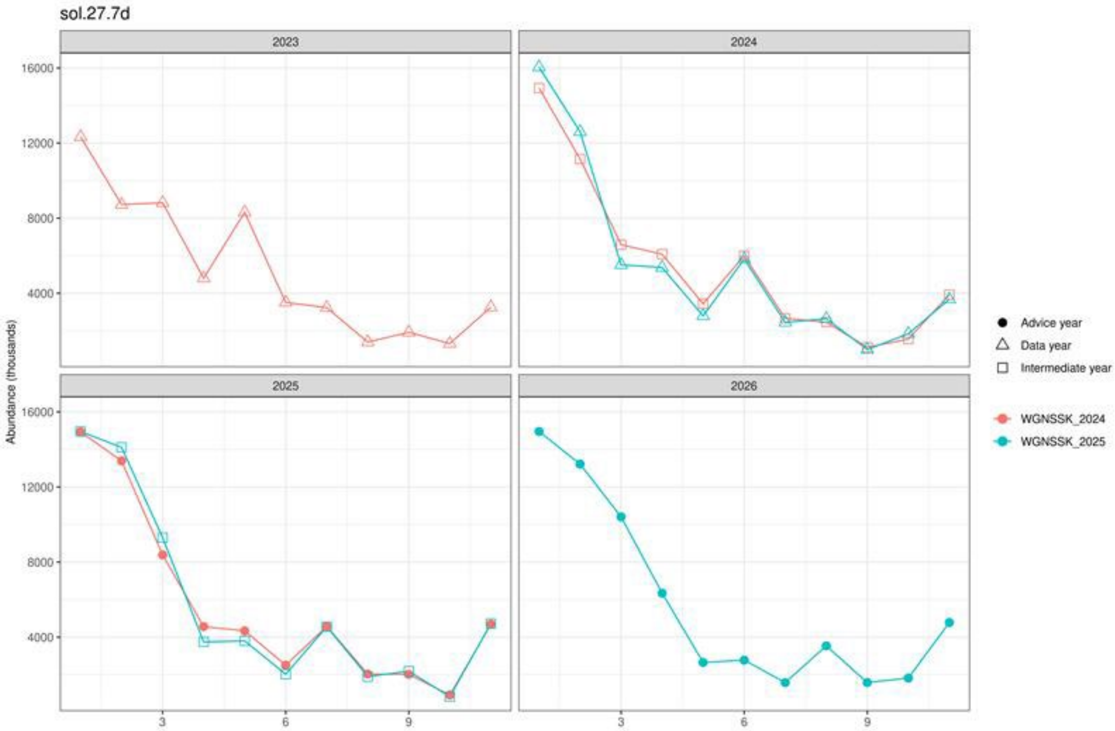


Figure 17.38. Sole in Division 7.d. Comparison of last year's stock numbers at age (estimated, intermediate year and advice year forecast assumption) against the stock numbers at age (estimated, intermediate year and advice year forecast assumption) of this year's assessment.

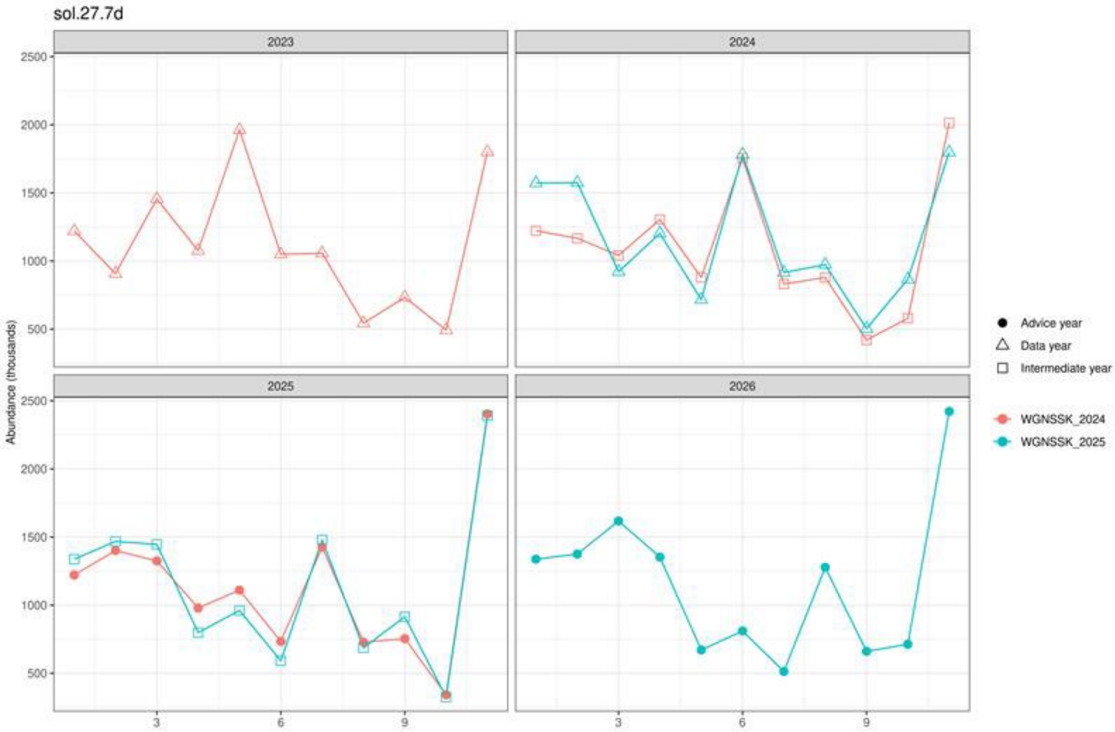


Figure 17.39. Sole in Division 7.d. Comparison of last year's biomass at age (estimated, intermediate year and advice year forecast assumption) against the biomass at age (estimated, intermediate year and advice year forecast assumption) of this year's assessment.

Contents